

Learning Coordinated Trajectories for Non-Stop Flying Carriers Holding a Cable-Suspended Load

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CHAPTER 1

Introduction

1.1 Motivation

Unmanned Aerial Vehicles (UAVs) have proved their great capability in transporting cable suspended loads in recent years. They offer the ability to reach places that are dangerous or remote and can transfer equipment that is otherwise difficult to move [1]. This can be of great utility during rescue operations, construction, humanitarian aid and logistics. While early efforts focused on single multirotors, they often face engineering challenges due to their limited payload capacity, endurance and ability to control the full load pose. To counter these challenges, collaborative teams of multirotors have been proposed.

Multirotors were mainly used because they are more agile than fixed-wings carriers and can hover. However, these solutions tackle only the limited payload capacity and the full pose control problems. They still suffer from reduced operational range due to their lower energy efficiency. On the other hand, fixed-wings platforms offer better energy efficiency [2] and thus longer operational range and higher cruising speeds. Hence, fixed-wings platforms are better suited for applications that require long range. This comes at the cost of a more complex control architecture because fixed-wings carriers cannot simply hover in place.

Multiple fixed-wings UAVs need to coordinate among themselves in order to allow the payload to stay stationary while they are loitering. This coordination is also needed in many other scenarios in the trajectory control of the payload. For this reason, most previous efforts were using multirotors. Recent works by [3],[4] and [5] showed that it is possible to calculate coordinated trajectories for the carriers online, such trajectories would respect the non-stopping condition while also maintaining load tracking. The solution however, was based on a centralized controller that calculated all the trajectories and then assigns each carrier its trajectory. Additionally, to calculate the trajectories online it had to solve an NLP each time step. This makes onboard deploying of the solution particularly challenging due to the high computational cost.

In this work, we propose a decentralized solution based on Imitation Learning (IL) and Multi-Agent Reinforcement learning (MARL). The proposed solution also allows for onboard deployment as the computational cost per step is greatly reduced to the inference time. Additionally,

its decentralized nature allows it to be deployed with great flexibility in many applications.

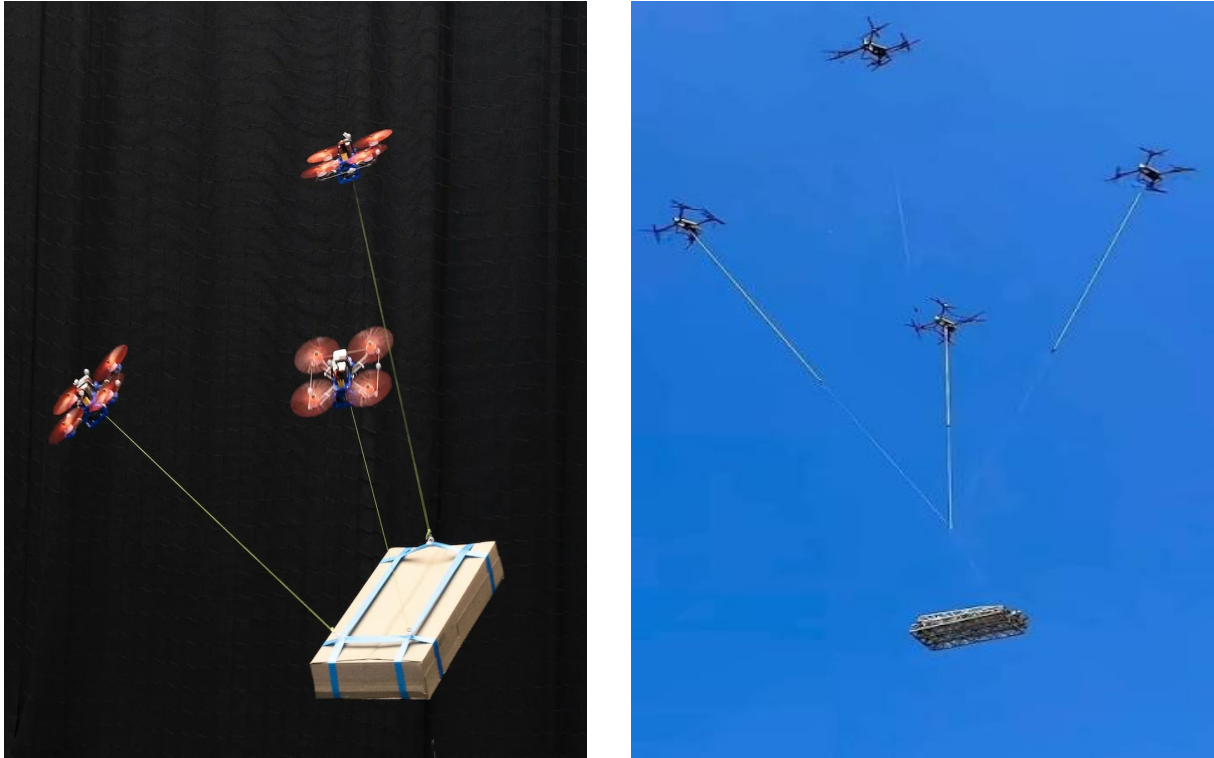


Figure 1.1: Cooperative teams of multirotor UAVs transporting a cable-suspended payload.

1.2 Main challenges

Fixed-wings UAVs present a unique and underexplored challenge in that they cannot hover. Since they always have to be moving, that means they will apply additional forces and torques on the payload other than the desired ones to move it along its trajectory. The main idea to solve this problem is to have these additional forces (referred to as internal forces) cancel each other out such that only the desired wrench (forces and torques) on the payload remains.

Hence, coordination between the carriers is a requirement. However, this coordination can still be achieved through well designed decentralized mechanisms depending on the load state and other local observations, without the explicit need for inter-carrier communication or measurements.

Fixed-wings UAVs also present a subtle but nontrivial problem, since the carrier needs to be always moving, the load state to carrier action map is not a well-defined function. In other words, we can have the same load state but different carrier actions in different times. Consequently, the carrier needs to take decisions based on memory or some understanding of time. Moreover, the actions of the carriers are not identical at the same moment in time and at the same payload state. This is due to the fact that the internal forces that will cancel each other need to be produced by different carrier trajectories.

Furthermore, since the policy is decentralized, each carrier will have to make its own ob-

servation of the load. We can imagine using an AprilTag on the load and each carrier has its own camera that sees it. Hence, each observation is going to be subject to its own noise and delay, leading to desynchronization in the load state between the different carriers. Therefore, the policy needs to learn good actions with limited and potentially outdated information. This naturally leads to a Partially Observable Markov Decision Process (POMDP) problem formulation. Reinforcement Learning (RL) provides a natural framework for learning control policies in such uncertain and partially observable environments. Finally, the problem has many other aspects that are beyond the scope of this thesis (e.g. enforcing a minimum turning radius).

The goal of this thesis is to answer whether a fully decentralized policy (using a combination of learning-based and classical methods) is able to coordinate the carrier trajectories even when they do not fully agree on the state of the system. Instead of relying on a centralized controller that has access to the full and true state, can they still learn a reactive policy that is able to stabilize the load and maintain their coordinated trajectories? Taking into consideration the carriers have their own local heterogeneous asynchronous observations. Additionally, the policy must be lightweight enough to deploy onboard and execute in real time.

CHAPTER 2

Literature review and Project Contribution

Cable-suspended cooperative payload transportation using UAVs has garnered considerable attention in recent years. Researchers have explored a wide range of techniques, from classical methods such as Optimal Control [6] and closed-loop approaches with carrier trajectories calculated online using a Nonlinear Programming (NLP) solver, to learning-based methods such as Imitation Learning and Reinforcement Learning [7] [8] [9], with each offering its own advantages and disadvantages.

2.1 Multi-rotor cooperative transportation

In [6], the authors propose a centralized kinodynamic motion planner that formulates a finite-horizon Optimal Control Problem (OCP) and solves it online in a receding-horizon fashion. The approach aims to enable high-speed, high-acceleration manipulation of a cable-suspended payload while accounting for the strong dynamic coupling between the UAVs, cables, and payload. While it achieves strong results, it relies on a centralized kinodynamic planner, which may pose challenges for fully onboard deployment of the complete control system.

In [7], a centralized kinodynamic planner is distilled using DAgger-based imitation learning into a homogeneous, decentralized UAV policy¹. This allows the approach to achieve agile full-pose cooperative manipulation using only local observations and without communication. While the policy was decentralized, it was not deployed onboard during the actual testing due to its high computational cost.

In [8], a decentralized cooperative control policy was learnt directly through multi-agent reinforcement learning (MARL), trading explicit trajectory-planning capability for much cheaper computational cost and greater robustness to heterogeneous agents and agent failure. Additionally, the policy was only tasked to learn high level commands/action such as reference acceler-

¹**Distillation** is a form of imitation learning that involves transferring knowledge from a teacher policy, typically obtained through planning or expert demonstration, to a smaller or more deployable student policy by training the latter to reproduce the teacher's actions. **DAgger (Dataset Aggregation)** is an imitation-learning algorithm that iteratively collects expert actions on states visited by the learned policy and aggregates these samples into the training dataset, thereby reducing the distribution-shift problem encountered when the learner visits states not represented in the original demonstrations.

ation. The action is then handed to a low level controller that executes the actual tracking. In this case, since the policy had dramatically lower computational cost, actual onboard execution was possible. The policy did not rely on any inter-drone communication or measurements and relied on observations such as the load state as a way of implicit communication.

In [9], instead of relying on a low level controller, an end-to-end decentralized cooperative control policy was learnt using RL. The policy takes as input the sensor measurements and outputs the Pulse-width modulation (PWM) directly to drones' motors. This eliminates the need for a low level controller and simplifies the deployment. However, while the policy did not rely on explicit communication between the drones, it relied on inter-drone measurements, with each drone measuring the relative position between itself and the other drones. Such measurements were subjected to noise during the training to make the policy resilient during deployment in real life scenarios. This is in contrast to [8] where the measurements were in general noise-free but did not employ any type of inter-drone communication or measurements.

2.2 Fixed wings cooperative transportation

However, all of these works have focused on multirotor UAVs, while comparatively little attention has been given to fixed-wing UAVs. More recently, a new line of research has emerged exploring the same problem using fixed-wing UAVs [3] [4] [5]. In [3], the existence of static equilibria for a cable-suspended load transported by non-stopping flying carriers is investigated. The study shows that fixed-wing carriers can maintain a fixed pose for a cable-suspended load using at least three carriers, with some exceptions and degenerate cases. The work presents a theoretical feasibility study supported by numerical simulations in MATLAB/Simulink. However, only steady-state equilibrium is considered, without addressing the transient behavior of the system. Furthermore, the carrier reference trajectories are precalculated offline and executed in open loop, meaning that there is no reaction to disturbances, system uncertainties, or obstacles, and perfect coordination between the carriers is assumed.

In [4], an algorithm for generating coordinated non-stop trajectories for three or more carriers holding a cable-suspended load is presented. The approach uses Hamiltonian cycles and graph theory to generate the coordinated trajectories, with the results demonstrated through numerical simulations in MATLAB/Simulink. Similar to [3], only steady-state equilibrium is considered, without addressing the transient behavior of the system. Furthermore, the carrier reference trajectories are precalculated and there is no closed-loop control, meaning that the trajectories cannot react to disturbances or system uncertainties during operation.

In [5], a closed-loop control approach for cooperative cable-suspended payload manipulation using non-stopping flying carriers is presented. The approach considers a time-varying load trajectory, which can also include stopping, and consists of an outer-loop load-wrench controller, an inner-loop carrier-trajectory generator, and an optimization layer for the internal forces to enforce a strictly positive carrier velocity norm. The approach is experimentally validated using

Crazyflie quadrotors and the Crazyswarm software, with a Vicon motion capture system, in addition to numerical simulations in MATLAB/Simulink. However, the controller is centralized, is not deployed onboard, and assumes perfect knowledge of the system. Furthermore, the trajectories are calculated using a centralized controller together with a Nonlinear Programming (NLP) solver/optimizer, resulting in a high computational cost.

The works discussed above that tackled the non-stopping carriers problems all relied on calculating the UAVs trajectories centrally using classical control techniques. While they demonstrated several effective approaches to cooperative payload transportation using non-stopping carriers, they also highlighted a gap in using learning-based methods and decentralized strategies to tackle the same problem. To the best of the author's knowledge, this research direction was not explored before. Motivated by this gap, this work explores the use of Imitation Learning and Reinforcement learning as a potential research direction.

CHAPTER 3

System Modelling and Problem formulation

The most straightforward way to test the decentralization hypothesis is to build upon the controller and optimizer introduced in [5]. We will refer to these controller and optimizer as the Classical Control Stack (CCS). Instead of forcing the policy to learn everything from scratch, we can rely on the CCS since it can already fly and generate trajectories in a centralized fashion. We just need to make the policy learn to mimic, correct and augment its behavior under loss and reconstruction of some information.

The structure of the CCS lends itself easily to decentralization, hence, we will exploit this property and use parts of it and its architecture. This will be discussed in detail in later sections, for now we start by defining the problem, how the classical control pipeline solves it and what will be our modifications to decentralize it. The system model and the CCS were mainly developed in [3], [4] and [5].

3.1 System Model

Consider a system of $n \geq 3$ non-stopping rigid carriers collaboratively carrying a rigid payload using cables. The carriers are required to maintain a strictly positive velocity to avoid stalling. The connection positions between the cables and the payload are known. Refer to Figure 3.1 for a schematic representation of the considered scenario.

Remark: In [4], it is established that a minimum of three non-stopping carriers are needed to maintain a cable suspended load static while allowing for each carrier to maintain a loitering positive velocity norm.

Modelling assumptions:

- The cables are massless and inextensible
- The cables are taut and can only transmit tensile forces
- The carriers and the load are rigid bodies

Let the inertial world frame $\mathcal{F}_W = \{O_W, (x_W, y_W, z_W)\}$, with gravity acting along the opposed direction of the z_W -axis, and let the payload's body-fixed frame $\mathcal{F}_L$ be attached to the payload's center of mass (CoM). Let the payload mass be $m_L \in \mathbb{R}_{>0}$ and inertia matrix $J_L \in \mathbb{R}^{3 \times 3}$. Let the i -th, $i \in \{1, \dots, n\}$, carrier mass be $m_i \in \mathbb{R}_{>0}$. The i -th carrier is connected to a fixed attachment point B_i on the payload via a cable of constant length $L_i \in \mathbb{R}_{>0}$, with unit direction vector $q_i(t) \in \mathbb{S}^2$ pointing from the attachment point to the carrier (representing the cable direction in the inertial frame).

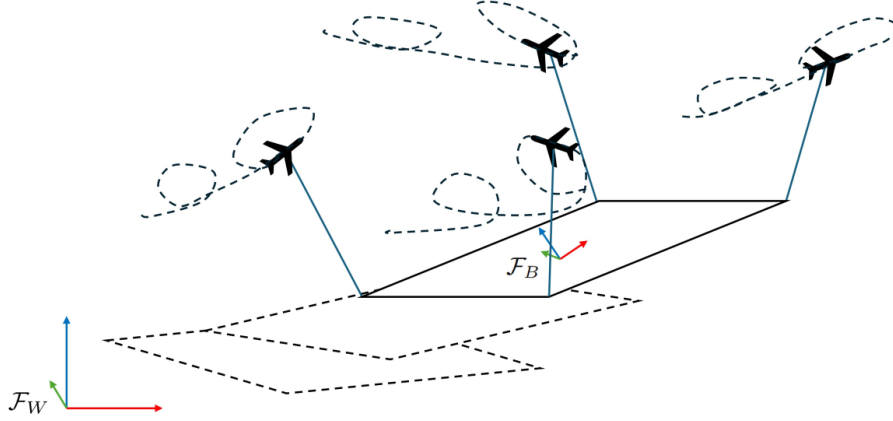


Figure 3.1: Schematic representation of n non-stopping fixed-wing carriers connected via cables to a rigid payload, showing the inertial world frame $\mathcal{F}_W$ and the payload's body-fixed frame $\mathcal{F}_B$. Reproduced from [5].

We can then calculate the position of the i -th carrier $p_{R_i}(t) \in \mathbb{R}^3$ expressed in $\mathcal{F}_W$ using:

$$p_{R_i}(t) = p_L(t) + R_L(t) {}^B b_i + L_i q_i(t) \quad (3.1)$$

where $p_L(t) \in \mathbb{R}^3$ and $R_L(t) \in SO(3)$ denote the payload position and orientation in $\mathcal{F}_W$, and ${}^B b_i \in \mathbb{R}^3$ represent the cable attachment points expressed in $\mathcal{F}_L$. By differentiating (3.1) we get

$$\dot{p}_{R_i}(t) = \dot{p}_L(t) + \dot{R}_L(t) {}^B b_i + L_i \dot{q}_i(t) \quad (3.2)$$

with $\dot{R}_L(t) = R_L(t)S(\omega_L(t))$ where $\omega_L(t) \in \mathbb{R}^3$ is the payload angular velocity w.r.t. $\mathcal{F}_W$, expressed in $\mathcal{F}_L$, and $S(\cdot)$ is the skew-symmetric operator

Equations (3.1) and (3.2) show that the carriers' position and velocity are inherently constrained by the load state and the geometrical configuration of the attachment point and the cable.

This means that the forces are the real command from whatever control strategy we design, and determining the desired position and velocity is a matter of solving the kinematic equations

(3.1) and (3.2). This is also relevant for choosing the action space of the policy. Instead of trying to output position/velocity/acceleration set points for the drones, we will output forces directly since this is the true independent variable that determine the carriers' and the payload's state. In particular, $q_i(t)$ and $\dot{q}_i(t)$ control the carriers' state. While they can be seen as the cable direction and its derivative, they also serve as unit vectors defining the directions of the forces.

$$f_i(t) = T_i(t)q_i(t) \quad (3.3)$$

where $T_i(t) \in \mathbb{R}_{\geq 0}$ is the tension in the i -th cable and it generates a force $f_i(t) \in \mathbb{R}^3$ on the payload along the direction q_i .

The payload dynamics follow the Newton–Euler equations:

$$m_L \ddot{p}_L(t) = -m_L g e_3 + \sum_{i=1}^n f_i(t), \quad (3.4)$$

$$J_L \dot{\omega}_L(t) = -\omega_L(t) \times J_L \omega_L(t) + \sum_{i=1}^n S({}^B b_i) R_L(t)^\top f_i(t). \quad (3.5)$$

Let $f(t) = [f_1(t)^\top \dots f_n(t)^\top]^\top \in \mathbb{R}^{3n}$ be the collection of the cable forces. And let $w(t) = [f_L(t)^\top \tau_L(t)^\top]^\top \in \mathbb{R}^6$ representing the wrench at the load CoM, where $f_L(t) \in \mathbb{R}^3$ and $\tau_L(t) \in \mathbb{R}^3$ denote the force and the moment components, respectively. The relation between these vectors is given by

$$w(t) = G(R_L(t))f(t), \quad (3.6)$$

where $G(R_L(t)) \in \mathbb{R}^{6 \times 3n}$ is the grasp matrix defined as

$$G(R_L(t)) = \begin{bmatrix} I_3 & \cdots & I_3 \\ S({}^B b_1)R_L(t)^\top & \cdots & S({}^B b_n)R_L(t)^\top \end{bmatrix}. \quad (3.7)$$

It is easy to see that the first three rows map the forces f to translational force $f_L(t)$. On the other hand, the last three rows map them to the moment $\tau_L(t)$ about the payload CoM. We can see that this map depends on the load orientation R_L and the fixed attachment points ${}^B b_i$ in the body frame.

This formulation enables efficient computation of the cable forces required to generate a desired load wrench, while providing the basis for internal force allocation and coordinated control of the carriers.

Indeed, we have that

$$f(t) = \underbrace{G(R_L(t))^{\dagger}w(t)}_{=: f_g(t)} + \underbrace{N(R_L(t))\lambda(t)}_{=: f_{\lambda}(t)}, \quad (3.8)$$

where $(\cdot)^{\dagger}$ indicates the right pseudo inverse and $N(R_L(t))$ is a matrix whose columns identify a basis of the nullspace of $G(R_L(t))$.

Assuming that the attachment points on the load are well designed, we can guarantee that $\text{rank}(G(R_L(t))) = 6$. Consequently, dimension of $\text{null}(G(R_L(t)))$ is $3n - 6$ which is equal to the number of columns of $N(R_L(t))$ and the number of independent internal forces component. This number of independent internal force components defines how many degrees of redundancy are available in the system. This redundancy allows for internal forces $f_{\lambda}(t) \in \mathbb{R}^{3n}$ belonging to $\text{null}(G(R_L(t)))$ to redistribute the cable tension forces without changing the applied wrench on the load and by extension its motion. Since this work doesn't go into the design of the classical control pipeline, the choice of $N(R_L(t))$ to parameterize these internal forces by a set $\lambda(t)$ of n functions was used as is from [4].

As a consequence of this architecture, the load trajectory tracking is cleanly isolated from the carrier trajectory feasibility (i.e. maintaining the velocity norms of all the carriers above a certain threshold $\varepsilon > 0$).

$$\|\dot{p}_{R_i}(t)\| \geq \varepsilon > 0. \quad (3.9)$$

Precisely, the load tracking will be completely achieved by controlling $f_g(t)$ while, the carrier trajectory feasibility will be done by controlling the internal forces $f_{\lambda}(t)$.

3.2 Classical Control Stack (CCS)

3.2.1 Load tracking

The load tracking is achieved by calculating the desired wrench, the forces necessary to apply this wrench can then be calculated using the grasp matrix $G(R_L(t))$ using equation (3.6). To calculate the desired wrench, simple PID controllers can be used (one for the forces and one for the torques). But first the error signal must be provided to these controllers. The desired load trajectory is characterized by the desired position, orientation, and corresponding velocities over time, denoted by $p_{L,d}(t) \in \mathbb{R}^3$, $R_{L,d}(t) \in SO(3)$, $\dot{p}_{L,d}(t)$, and $\omega_{L,d}(t) \in \mathbb{R}^3$, respectively. The pose and velocity errors of the load relative to this desired trajectory are defined as follows:

$$e_p(t) = p_L(t) - p_{L,d}(t) \quad (3.10)$$

$$e_R(t) = \frac{1}{2} (R_{L,d}(t)^\top R_L(t) - R_L(t)^\top R_{L,d}(t))^\vee \quad (3.11)$$

$$e_v(t) = \dot{p}_L(t) - \dot{p}_{L,d}(t) \quad (3.12)$$

$$e_\omega(t) = \omega_{L,d}(t) - \omega_L(t) \quad (3.13)$$

Where $(\cdot)^\vee$ is the inverse of the skew-symmetric operator

Given these errors, the desired load wrench $w_d(t) = [f_{L,d}(t)^\top \tau_{L,d}(t)^\top]^\top$ can be calculated using the PID controllers as follows:

$$f_{L,d}(t) = m_L g e_3 - K_p e_p(t) - K_v e_v(t) - K_i \int_0^t e_p(\tau) d\tau, \quad (3.14)$$

$$\tau_{L,d}(t) = \omega_L(t) \times J_L \omega_L(t) - K_R e_R(t) - K_\omega e_\omega(t) - K_{iR} \int_0^t e_R(\tau) d\tau, \quad (3.15)$$

with tunable gains $K_p, K_v, K_i, K_R, K_\omega, K_{iR} \in \mathbb{R}^{3 \times 3}$

3.2.2 Carrier Trajectory / Internal Force Calculation: Optimization layer

By construction, the internal cable force components $f_\lambda(t)$ in (3.8) do not have an effect on the wrench applied to the load CoM. In [5], a method of controlling these forces online is formulated. An optimization problem is solved at each time step to keep the carrier velocities above a specified threshold.

While we do not intend to use this optimizer in the solution presented in this work, its structure is still relevant because we will attempt to mimic its behavior (we will expand on this point in a later section). Consequently, it is worth to briefly cover it.

Given $w_d(t)$ based on (3.14)-(3.15), and the corresponding cable forces $f(t) \in \mathbb{R}^{3n}$ generating such a wrench is:

$$f(t) = G(t)^\dagger w_d(t) + N(t) \lambda(t), \quad (3.16)$$

where $\lambda(t)$ represents a functional variable to be optimized, and we use the short hand $G(t) = G(R_L(t))$ and $N(t) = N(R_L(t))$.

The choice of $\lambda(t)$ and design of $N(t)$ was extensively studied in [4] and [5]. In this work, it is sufficient for us to just take them as inputs. $\lambda(t)$ is of particular interest, since it will be the main lever to shape the internal forces of the carriers. It is chosen to be:

$$\lambda_i(\xi(t), A(t), t) = A(t) \cos(\xi(t)t + \phi_i) \quad (3.17)$$

Where the parameters $\phi_i \in [-\pi, \pi)$, $i = 1, \dots, n$ represent the constant phase shifts of the carriers (not necessarily distinct but some have to be). Note that this choice of the phase shifts only works for an even number of carriers[4].

Whereas the pair $x(t) = (\xi(t), A(t))$ represent the magnitude and the frequency and are optimized online to enforce a desired lower-bound on the norm of the carrier velocities as in eq. (3.9) (they are shared among the carriers).

Moreover, it can be shown that the force derivative (required to calculate $\dot{q}_{i,d}(t)$ and hence $\dot{p}_{Ri,d}(t)$) can be calculated analytically and numerically from the previous equations.

The optimization problem solved online is

$$x^*(t) = \arg \min_{x(t) \in \mathbb{R}^2} J(x(t)) \quad \text{s.t.} \quad \text{eq. (3.9)} \quad \forall i = 1, \dots, n, \quad (3.18)$$

with the objective function

$$\begin{aligned} J(x(t)) = & (\xi(t) - \xi(t^-))^2 + (A(t) - A(t^-))^2 \\ & + w_{\text{pos}} \|\lambda(x(t)) - \lambda(x(t^-))\|_2^2 + w_{\text{vel}} \|\dot{\lambda}(x(t)) - \dot{\lambda}(x(t^-))\|_2^2. \end{aligned} \quad (3.19)$$

where t^- denotes the previous time instant, and w_{pos} and w_{vel} weighting factors (in this implementation they were observed to be not helpful therefore set to zero). The cost function is designed to penalize large instantaneous variations in the cable forces, thereby reducing abrupt changes in the carriers' trajectories. Solving the optimization problem yields the optimal coefficients of the internal force function, $\lambda_i(\xi^*(t), A^*(t), t) \equiv \lambda_i^*(t)$, for $i = 1, \dots, n$.

In summary, the load state and its reference are used to calculate the desired wrench w_d . Additionally, the load state is also used to calculate $G(t)$ and $N(t)$. Finally, the optimizer calculates $\lambda(t)$. They are combined as in eq. (3.8) to generate the desired force which is then used in eq. (3.1) and (3.3) to calculate the desired position. The desired velocity is calculated from the desired force derivative which can be calculated in a similar albeit more involved manner. The desired positions and velocities are then passed to the low level controller of each drone.

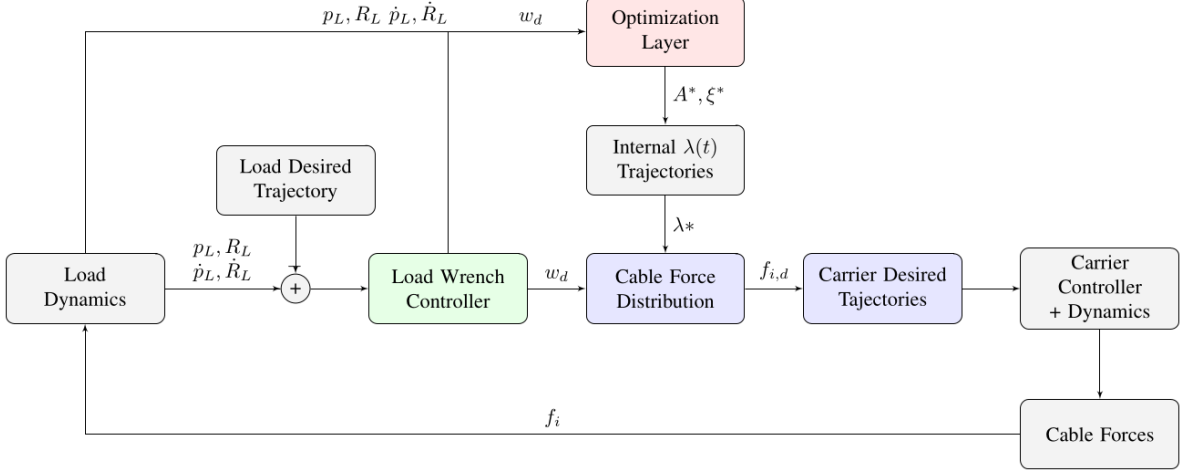


Figure 3.2: Overall classical control pipeline scheme, including the load-wrench controller (green block), the carrier-trajectory generator (blue blocks), and the optimization layer (red block). Reproduced from [5].

3.3 Learning-based methods

This section contains the main problem formulation of the contribution of this work. To achieve that, we split the work in two phases. Phase one will mainly target eliminating the optimizer since it takes the bulk of the computational time per step and therefore it is the biggest obstacle to onboard deployment. Phase two will focus on modifying the architecture and using Multi Agent Reinforcement Learning (MARL) to make it fully decentralized (no communication between the carriers with heterogeneous asynchronous observations).

Phase One: Imitation Learning(IL)/Behavioral Cloning(BC)

Objective of Phase One

Since the optimizer was already shown to work well, the proposed solution in this phase is to distill its behavior into a student policy using IL. The goal would be to train a Neural Network (NN) to mimic the output of the optimizer, eliminating the need for using it.

This dramatically reduces the computational cost allowing the trajectories to be calculated online if deployed in real time and also lays the ground work for Phase Two. Additionally, this makes deploying or training the system entirely on a GPU much more feasible, since calculating $\lambda(t)$ in the CCS relies on standard NLP formulations using tools such as CasADi and solvers such as IPOPT, which rely on CPU-based computation. In contrast, a NN can be trained and used for inference on a GPU, making the calculation of $\lambda(t)$ GPU-based and enabling parallelization.

This phase does not handle decentralization but just focuses on replacing the computationally expensive optimizer with a cheaper policy without changing the architecture. So the student policy is trained and deployed centrally with access to clean global states.

Action Space: $\lambda(t) \in \mathbb{R}^n$

- Keep wrench controller and nullspace matrix

$$f(t) = f_g(t) + \underline{f_\lambda(t)} \quad (3.20)$$

$$f_\lambda(t) = N(H, t) \cdot \underline{\lambda(t)} \quad (3.21)$$

- Load tracking guaranteed by construction ($G \cdot N = 0$)
- Doesn't handle desynchronized or heterogeneous observations

We face an interesting choice here, either to train the NN to output $\lambda(t)$ directly or output just $\xi(t)$, $A(t)$ and we can then reconstruct $\lambda(t)$ using the time t and phase shifts ϕ_i using eq. (3.17). Outputting $\xi(t)$ and $A(t)$ would be easier since they are not as dynamic as $\lambda(t)$ and don't have a periodic nature. However, in this work we chose to learn $\lambda(t)$ since this gives the NN control over the internal forces shape. The NN behavior can then be modified and fine-tuned using Reinforcement Learning (RL) to achieve certain requirements. A possible extension would be shaping the reward function to account for fixed-wings dynamics other than the non-stopping constraint (e.g. minimum turning radius). Outputting $\xi(t)$ and $A(t)$ would lock us into using eq. (3.17) with the frequency and amplitude being our only two degrees of freedom.

Phase two: Residual MARL

Objective of Phase Two

The goal of phase two is to move from one centralized controller that calculates $f_g(t)$ and one centralized student policy that calculates $f_\lambda(t)$ (from Phase One) into a decentralized controller and policy. For brevity, the controller and student policy will be later referred to together as the base controller, as they provide the foundational control structure of the final architecture. The residual RL policy is subsequently added on top of this base controller to form the complete architecture.

In this setup, each carrier takes its own measurements (potentially noisy and desynchronized) and uses local modified base controller (modified to run decentrally) and its local residual policy to calculate its own action. This will require modifying the architecture and using RL to help each carrier take "good" actions under this loss of information. The local instances of the base controller and residual policy will be homogeneous (i.e. they are all identical).

Action Space: $\delta f \in \mathbb{R}^{3n}$

- Keep wrench controller and use the NN that behaviorally clones the optimizer.
- Run decentralized

$$f(t) = f_g(t) + f_\lambda(t) + \underline{\delta f(t)} \quad (3.22)$$

- Handles desynchronized or heterogeneous observations
- Uses the residual force to maintain sensible carrier trajectories and load trajectory tracking

Phase One (IL) was chosen to be implemented before Phase Two (MARL) because to train the MARL policy will require running the simulation environment along with the base controller for potentially billions of steps. This makes the training process extremely sensitive to the computational cost per step and therefore it is necessary to reduce it where possible.

CHAPTER 4

Methodology

4.1 General discussion on the decentralization of the controller

To decentralize the controller, our approach will be to make each carrier calculate the full force vector $f(t) \in \mathbb{R}^{3n}$ and then only apply its own slice $f_i \in \mathbb{R}^3$. This is the easiest way to make the least amount of modifications to the original centralized controller. However, in true decentralization, the carriers do not have access to the clean global information that the centralized controller has access to, thus they will need to rely on their local (potentially noisy and delayed) observations.

This means that decentralization will be done in two steps:

1. Deploying the base controller separately on each carrier while keeping global information intact and accessible to all of them
2. True decentralization where each carrier relies only on local observations (potentially noisy and delayed).

To investigate the first step, we decompose the force vector as:

$$f(t) = f_g(t) + f_\lambda(t) \quad (4.1)$$

To calculate the full force vector $f(t) \in \mathbb{R}^{3n}$ we need to calculate its two main components. Firstly, f_g is directly calculated from the load state and reference using a PID controller eq. (3.10) to eq. (3.15), so no global observations about the other drone states are required. This makes calculating it decentrally trivial.

Calculating f_λ and λ

f_λ is a little bit more complicated, because it relies on a combination of the load state, drone velocities, and the optimizer's internal state of A and ξ , a phase shift assigned to each drone and time. The load state is locally observable to each drone so it is not an issue.

The lambda vector comes directly from the optimizer/student policy. The optimizer depends on the drone velocities and its own memory of A , ξ and ϕ_i 's.

By recalling eq. (3.17), these are the main components used to reconstruct lambda along with time:

$$\lambda_i(\xi(t), A(t), t) = A(t) \cos(\xi(t)t + \phi_i) \quad (4.2)$$

The phase shifts ϕ_i 's are constant for a given configuration so the policy can just memorize it for each lambda. A and ξ are more intricate because they depend on the drone velocities. Precisely, any drop in the velocity of any drone leads to the change of A and ξ for the whole formation.

All the carrier velocities are needed due to the nature of calculating f_λ . Since $f_\lambda(t) = N(H, t) \cdot \lambda(t)$ (eq. (3.21)), the whole lambda vector is needed even just to calculate the f_{λ_i} slice of one drone. This depends on the structure of matrix N but for the purposes of this thesis it is enough to say that it is not diagonal. Therefore, f_{λ_i} is not only dependent on λ_i , but also on other λ_j 's, where $j \neq i$.

The other carrier velocities, $\dot{p}_{R_j}(t)$, and the forces applied by them, f_j , calculated from f_{g_j} and f_{λ_j} , where f_{λ_j} is in turn calculated from λ_j , $j \neq i$, are the global information that will be corrupted by decentralizing the controller. We will attempt to estimate them analytically, but these are only reliable estimates if all carriers agree in their measurements and estimates (which we will assume for Phase One). Under realistic decentralization conditions, this assumption will not hold, and this is where Phase Two and the use of MARL come in.

Choice of NN architecture

Making a neural network infer the internal states (A and ξ) is possible by using a smart design of the observation space if a Multi-Layer Perceptron (MLP) is used (since MLPs have no memory elements or internal states). Alternatively, we directly use a Gated Recurrent Unit (GRU), which has an internal state. In this work an MLP is chosen for phase one due to its simplicity and lightweight nature. To compensate for the lack of an internal state, multiple historical data points of specific features are stacked and fed together as input to the MLP, more on this will be discussed later.

Estimating other carrier velocities

An approximation of the drones' velocities can be analytically computed in a manner similar to what is done in [5]. This calculation uses as input: the observed load state ($p_L(t)$, $\dot{p}_L(t)$, $R_L(t)$, $\dot{R}_L(t)$), the commanded full force vector and its derivative ($f(t)$, $\dot{f}(t)$), the commanded wrench and $\lambda(t)$ and its derivative ($w_d(t)$, $\lambda(t)$, $\dot{\lambda}(t)$). The resulting velocities will remain reasonably accurate as long as the drones agree on the load state measurement and all start from the same

initial (A and ξ). For phase one, this agreement is assumed. These calculated velocities can then be fed into the MLP as a proxy for global information.

Time/Periodic Nature of $\lambda(t)$

Lastly, while the optimizer uses time explicitly, by examining eq. (4.2), we can see that it is not actually explicitly needed. What is actually needed is to time the periodic nature of the lambdas. The policy can learn this periodicity again using smart observation space design, by feeding multiple historical data points of specific features and by using time embedding techniques such as Fourier time encoding, this idea will be discussed in detail in a later section.

Student Policy

By employing all these techniques together, we can reasonably aim to train an MLP using Behavioral Cloning (BC) to learn mimicking the optimizer/expert. This MLP/student policy can then be deployed on each drone separately. It will output the full lambda vector, and then we calculate the full force vector using the classical pipeline math. Finally, each drone applies its own slice of the force vector, this requires that each drone knows its ID in order to know which slice to select. But this information can be communicated/agreed upon only once in the beginning.

This achieves two major advantages over the optimizer:

1. We cut the runtime dramatically.
 - 1.1 Optimizer takes $\sim 3\text{-}4$ ms/step (benchmarked on an Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz with 16GB of RAM) while a trained NN inference time takes $\ll 1\text{ms/step}$ on any modern GPU.¹
 - 1.2 Cutting the runtime is not only necessary for onboard deployment but also practically a requirement if we want to do Residual RL on top of the whole pipeline as in phase 2. In order to do RL we need to simulate billions of steps, therefore any extra computational cost per step quickly adds up. A single component in the pipeline taking 3-4 ms makes the RL practically infeasible.
2. We can do RL and warm start our actor from this behaviorally cloned policy so we can mold it to our needs (if it turns out to be helpful or required). As discussed earlier, this is enabled by our choice to output $\lambda(t)$ directly instead of (A and ξ) allowing us to fine tune and modify the student policy to satisfy other requirements. This was not pursued in the present work, but it could be considered in future work.

¹The theoretical compute time, based on FLOP count and peak GPU throughput, is on the order of microseconds. However, in practice, the total inference latency would be dominated by GPU overhead

4.2 Implementation details

The system model/plant that simulates the actual physics of the system was developed entirely inside Simulink. It was then exported as a Functional Mock-up Unit (FMU)² that python can step through. The optimizer was then developed on top of it in Python (using CasADi). The FMU can also be wrapped in a gymnasium environment to allow easy interface with PyTorch for IL and RL.

However, the FMU cannot be run on GPU directly so the number of parallel environments is limited by the CPU cores. This was chosen because the modeling of cable-suspended multi-body systems is susceptible to subtle formulation and implementation errors when performed manually. In contrast, Simulink provides a framework that facilitates the consistent and accurate formulation of such systems. Since the PID controller, the optimizer were going to be developed on top of it. A reliable ground-truth plant to validate them is necessary. Additionally, the main bottleneck in Phase One was running the CasADi optimizer that the net needs to imitate. The optimizer runs on the CPU and takes 3-4 ms/step, while the FMU alone takes <1 ms/step. More importantly, since the optimizer is CPU-bound and cannot be run in parallel on the GPU, there is no point in creating thousands of parallel plant environments, as we cannot create a corresponding number of parallel expert instances. Therefore, parallelizing the environment and running thousands of instances on the GPU would be of no benefit.

For phase 2 (the MARL phase), since the optimizer is replaced by a NN, the bottleneck shifted to simulating the environment/running the FMU. This marked the point at which the Simulink generated FMU choice became increasingly burdensome. However, it was still possible to train a prototype agent that demonstrates the feasibility of the proposed approach. For future work, the system model should be implemented within a GPU-parallelizable environment, such as JAX or NVIDIA Warp.

4.3 Phase One : Imitation Learning

Initial approach

As a starting point, we attempted a simple supervised learning setup. We generate a trajectory using the CCS (later referred to as the expert), we collect the inputs that we intend to feed the NN at each time step and finally we record the output λ calculated by the expert. This creates a Dataset $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N$, where our state s_i will be the input and action a_i is the λ output.

We can then train the policy neural network in a supervised learning setting, where regression

²An FMU (Functional Mock-up Unit) is a self-contained software package that contains the compiled code and necessary information to execute a simulation model according to the Functional Mock-up Interface (FMI) standard. It can therefore be thought of as a compiled version of the model, similar to C code that can be executed without requiring Simulink itself. In this work, the FMU is generated from Simulink and can be executed by the Python environment for simulation and interaction with the RL agent.

is performed on expert state–action pairs. The network architecture is a 4-layer MLP of size [28, 256, 256, 4].

As a first attempt, only one load trajectory (scenario) was used to simplify computational load and allow us to investigate whether the policy neural network (an MLP) can provide λ and fly the carriers successfully in closed loop. The main challenge is that during closed loop deployment, the policy receives states generated by its actions as inputs, rather than the expert-generated states it has seen during training. The initial trajectory was chosen with care to ensure that the optimizer is engaged multiple times during the trajectory so that $\xi(t)$ and $A(t)$ meaningfully change. This is to ensure that the net captures a broad range of the optimizer’s behavior.

After the initial attempt proved successful and the policy was able to closely mimic the optimizer, the approach was generalized to a diverse set of trajectories, as will be discussed in following sections.

4.3.1 State/Input Design

As discussed earlier, this is a critical design choice since the optimizer has many internal variables and the MLP has no memory element. Furthermore, λ is not just a function of these internal states but also a function of time. Indeed, given fixed ϕ , ξ and A , λ is still time varying since it is a sinusoid eq. (3.17).

We can generally split the variables that λ depends on to three main categories.

1. Physical system states: these are either measured or estimated or a combination of both. For us, these are the carrier velocities.
2. Internal variables: these are stored inside the optimizer and can either be fixed or variable depending on the physical system states and their past values. For us, these are ϕ_i , $\xi(t)$ and $A(t)$.
3. Time: in order to build the sinusoid after determining all the internal variables.

We discuss each of them in turn in the remainder of this subsection.

Physical system states

Naturally, the reconstructed velocities of the carriers $\dot{p}_{R_i}$, $i = 1, \dots, n$ are a going to be part of the state. For our particular implementation we chose $n = 4$, so the first part of our state is $\dot{p}_{R_{1,2,3,4}} \in \mathbb{R}^{12}$ i.e. the carrier velocity vectors stacked. While technically only the velocity norms are needed, the whole vectors were provided thus increasing the information available to the MLP and allowing it to calculate the norm internally.

Internal variables

The phase shifts ϕ_i , $i = 1, 2, 3, 4$ are constant for a given configuration and therefore can be

memorized. On the other hand, $\xi(t)$ and $A(t)$ are variable and depend on their previous values and the carrier velocity norms. Since the initial value for them in the optimizer is constant, a reasonable assumption would be that the network can memorize them if they remain unchanged. But since they only change based on the carrier velocities (which are provided to the policy), the policy can just memorize the initial condition and learn how/when to change them similar to the optimizer conditioned on the velocity norms. The challenge is MLPs don't have any memory or internal state. Consequently, if the policy learns to change $\xi(t)$ and $A(t)$ in a given step based on its input and what the expert action was, by the next step it will have no memory of this and will fall back to the default/initial $\xi(t)$ and $A(t)$ that structurally memorized. A reasonable remedy would be to supply the policy's last action as part of the current state/input. So $a(t) \in s(t+1)$ or in our case $\lambda(t) \in s(t+1)$.

For learning a single trajectory, this proved to be sufficient. However, during generalization it was not enough and the policy was not able to accurately mimic the optimizer. By analyzing its failure mode and the underlying causes, more history information was determined to be necessary (will be discussed in detail in the results section). As a result, instead of just supplying the last λ as an input to the next time step, log-spaced point taps (λ at lags 1, 2, 4, 8, 16, 32, 64, 128, 192, 256 steps) were used.

$$\begin{aligned} &[\lambda(t-1), \lambda(t-2), \lambda(t-4), \lambda(t-8), \lambda(t-16), \\ &\lambda(t-32), \lambda(t-64), \lambda(t-128), \lambda(t-192), \lambda(t-256)] \in s(t+1) \end{aligned} \quad (4.3)$$

This technique means we need to provide multiple historical λ 's that did not exist at $t = 0$. To address this challenge, we assume that the initial $\xi(t)$ and $A(t)$ are agreed on once in the beginning (either by communication or just hardcoding). They are then used to construct the historical λ 's by extrapolation in the past knowing the fixed phase shifts of the configuration.

Time/Periodic Nature of $\lambda(t)$

Although providing the time directly to the NN may appear to be the most straightforward solution, this approach leads to several problems. First of all, during the initial stage when we are training on one trajectory, this risks that the network just memorizes which actions to do just based on time. This completely invalidates the test and compromises the investigation of whether the provided state is enough to make the actions learnable. Secondly, it creates generalization risks when we attempt to run episodes longer than those used in the training. Lastly, it creates normalization problems since we normalize our observation space before feeding it to the net. By using larger and larger timestamps we distort this normalization and destabilize the resulting observations.

The main goal of using the time to determine the current phase in the sinusoid function. Consequently, we need only to encode the position in the oscillation. It can be argued that the previous λ 's carry this information and therefore should be enough. However, this might be too

demanding as any small perturbation or noise in the output can cause the NN to deviate and think it is in a different phase at the next time step. The perturbed output is used as input at the next time step, creating a feedback loop in which the resulting deviation can propagate and compound over subsequent steps. As the policy continues to act on increasingly perturbed observations, the instability can become self-reinforcing, while simultaneously driving the physical system away from its intended state.

It is important to note that this information is not encoded in any way in the load position, as it can be completely static while λ is oscillating. Nevertheless, an interesting observation is that since each carrier is always orbiting based on λ , its position and velocity implicitly encode its phase within the cycle. So the policy does not necessarily need to have access to time. The phase of the periodic signal is recoverable from the observable state of the system. In this sense, the dynamics of the environment effectively integrate the periodic signal into the carrier's physical state, allowing the policy to infer the corresponding phase of λ directly from the observation rather than relying on an external clock. The main challenge with this approach, though, is that it suffers from the same sensitivity and instability as depending on previous λ 's. Once the carrier position and velocity are perturbed or noised in any way, the policy's inference of the phase can deviate in an unrecoverable way that is similar to what was described in the previous paragraph.

A promising solution would be to use the output of a periodic function in time. One possible function can be a saw tooth, but an even better solution can be using sine and cosine functions.

Fourier time encoding/ Fourier feature mapping

Fourier time encoding (also called trigonometric clock embedding, sinusoidal time embedding, or Fourier feature encoding) is a way of representing time or phase as a vector of periodic functions instead of a single scalar [10].

$$\phi(t) = [\sin(\omega t), \cos(\omega t)] \quad (4.4)$$

This anchors the observation in time, making any deviations recoverable. This observation is also immune to noise or perturbations. Sine and cosine are combined since sine alone (or cosine alone) cannot uniquely identify the phase.

$\sin(\theta) = \sin(\pi - \theta)$, so two different phases can produce the same value.

With both sine and cosine, each phase corresponds to a unique point on the circle/phase: $\theta \longrightarrow [\sin(\theta), \cos(\theta)]$.

Furthermore, instead of using just one frequency, we can use a set of frequencies $\{\omega_i\}_{i=1}^K$.

$$\phi(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \sin(\omega_2 t), \cos(\omega_2 t), \dots, \sin(\omega_K t), \cos(\omega_K t)] \quad (4.5)$$

This is because $\xi(t)$ can change from trajectory to another and within the same trajectory so the policy needs to learn to switch to a different frequency and time itself using it. Specifically, we use seven frequencies $\{1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0\}$ rad/s. These frequencies were selected based on observing $\xi(t)$ on a wide variety of trajectories. It was observed that $\xi(t)$ never dips below 1.5 rad/s and never goes above 3.0 rad/s, so that was the appropriate range to provide. A spacing of 0.25 rad/s was selected to create a densely spaced frequency set, providing a finer representation of temporal variations within our narrow frequency range while avoiding redundancy and excessive growth in the dimensionality of the input representation. This enables the network to better distinguish between closely spaced temporal frequencies without increasing the total number of frequencies unnecessarily.

To summarize, the observation space of the Imitation learning policy is:

- Fourier time embedding/features (eq. (4.5) with $K = 7$):

$$\phi(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \sin(\omega_2 t), \cos(\omega_2 t), \dots, \sin(\omega_7 t), \cos(\omega_7 t)]$$

- Reconstructed carrier velocities:

$$\dot{p}_{R_i}(t), \quad i = 1, 2, 3, 4$$

- Log-spaced point taps (λ at lags 1, 2, 4, 8, 16, 32, 64, 128, 192, 256 steps):

$$[\lambda(t-1), \lambda(t-2), \lambda(t-4), \lambda(t-8), \lambda(t-16), \\ \lambda(t-32), \lambda(t-64), \lambda(t-128), \lambda(t-192), \lambda(t-256)]$$

4.3.2 Dataset Aggregation (Dagger)

While normal BC provides a simple and effective way to learn a policy from expert demonstrations by treating imitation learning as a supervised learning problem. The key assumption, however, is that the states encountered by the learner during deployment will look similar to those seen in the expert demonstrations.

In practice, this assumption often breaks down. A small mistake by the learned policy can take the agent into a state that the expert never visited. Since the policy has not learned how to act in these unfamiliar states, its errors can compound, causing it to drift further away from the expert's trajectory. This is known as covariate shift or the distribution mismatch problem [11].

Dagger addresses this issue by making the learner interact with the environment and querying the expert for the correct action on the states that the learner actually encounters. These newly labeled states are then aggregated with the original demonstrations to retrain the policy [11].

Thus, while BC learns to imitate the expert on the expert's state distribution, DAgger trains the policy on the states it is likely to encounter itself. This makes DAgger particularly useful for reducing compounding errors and improving robustness when the learned policy deviates from the expert.

Moreover, we do not suddenly expose the policy to its completely off-distribution states. We want to stay near the good orbit while slowly letting the policy explore off it. To accomplish that, DAgger defines the rollout/data-collection policy as

$$\pi_i = \beta_i \pi^* + (1 - \beta_i) \hat{\pi}_i, \quad (4.6)$$

where:

- π^* = expert policy
- $\hat{\pi}_i$ = current learned policy
- β_i = mixing parameter β

This can also be written explicitly in terms of λ as,

$$\lambda_i = \beta_i \lambda_i^* + (1 - \beta_i) \hat{\lambda}_i, \quad (4.7)$$

where λ_i^* is the expert-generated λ and $\hat{\lambda}_i$ is the current learned policy's λ output.

Unlike the original DAgger formulation, where the expert and learner policies are selected stochastically according to the mixing probability β , we use β as an interpolation coefficient to directly combine their actions. Thus, the executed action is obtained as a weighted combination of the expert and learner actions.

This continuous interpolation is well suited to a control setting that operates in a continuous action space.

Additionally, at each DAgger iteration, β is gradually decreased from 1 to 0 (e.g., in decrements of 0.1), resulting in a smooth transition from expert-guided behavior to the learned policy. For each value of β , data is collected using the resulting mixture of expert and learned actions and subsequently added to the aggregated dataset. The policy network is then retrained from scratch on the entire aggregated dataset before proceeding to the next iteration.

In this implementation, the β schedule that was used is:

$$\beta \in [0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.15, 0.1, 0.08, 0.06, 0.04, 0.02, 0.0]$$

The smaller $\Delta\beta$ steps closer to the end are to make the transition to policy closed loop control as smooth as possible.

The DAGger iteration can then be summarized in three steps:

1. Collect experience using

$$\pi_i = \beta_i \pi^* + (1 - \beta_i) \hat{\pi}_i \quad (\text{eq. 4.6})$$

2. Aggregate the collected experience with the total dataset and train on it
3. Decrease β and repeat

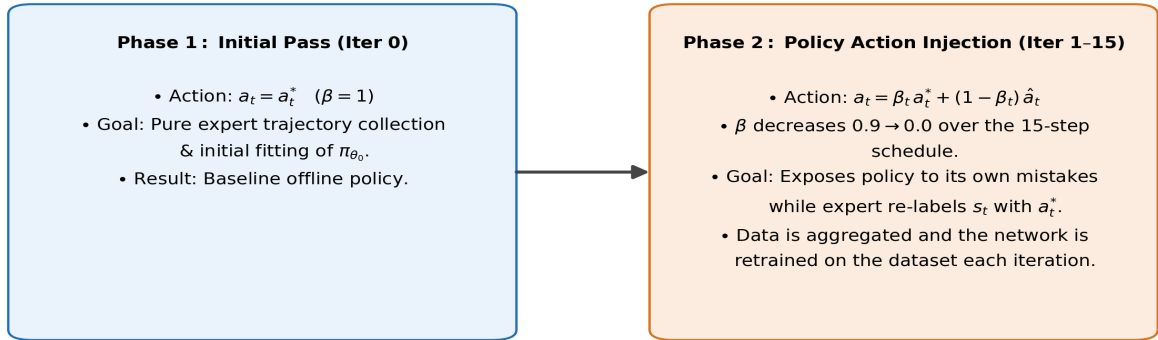
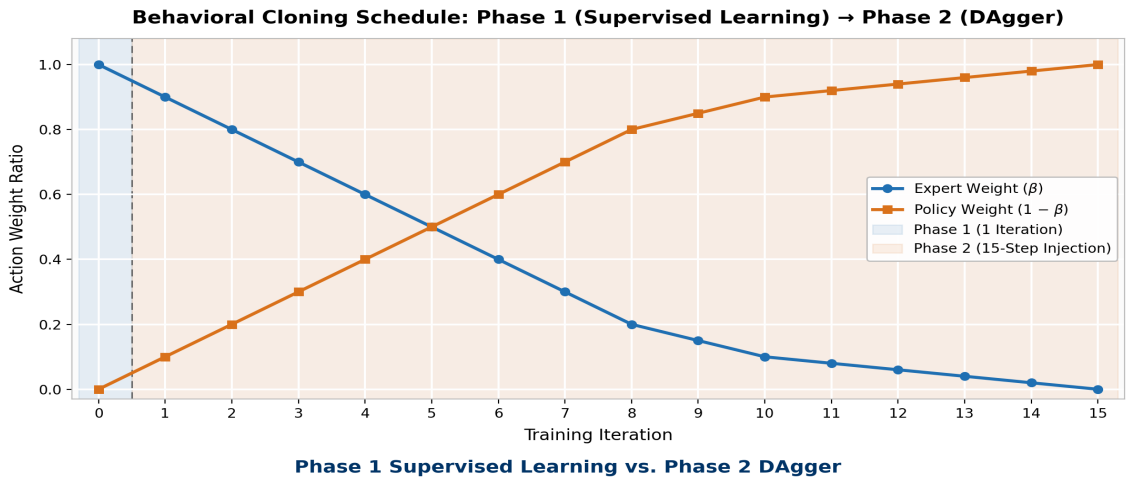


Figure 4.1: Behavioral Cloning Schedule: Phase 1 (Supervised Learning) → Phase 2 (DAGger).

Remark: The policy input $(\dot{p}_R, \lambda_{\text{taps}}(t - i))$, where $\lambda_{\text{taps}}(t - i)$ denotes the full set of log-spaced λ lags (eq. (4.3)), is reconstructed using the applied (mixed) λ history, rather than the policy output alone. This is necessary because $\lambda(t - i)$ represents a physical observation of the system state, specifically the cable force distribution that was actually applied to the system at each corresponding time step, rather than an internal memory state of the policy. Consequently,

the observation at time t must be consistent with the physical state generated by the applied action $\lambda_{\text{applied}}(t-i)$. Using the policy output alone could instead produce a physically inconsistent state in which the kinematic variables reflect the response to one cable-force distribution, while $\lambda(t-i)$ corresponds to another. Reconstructing the input from the applied λ history therefore ensures that the policy is trained and evaluated on the state distribution it actually encounters during execution.

This strategy worked well when training on a single trajectory, however, during its extension to a much larger set of trajectories to achieve generalization, it became increasingly inefficient due to two main reasons:

1. Training the MLP from scratch each iteration
2. Dataset size swelling

To tackle these challenges, we propose the following solutions.

Warm-Starting the MLP

The MLP is started from random weights at the beginning of each new DAgger iteration. This wastes many epochs to relearn stuff that can already exist in the old MLP weights. To counteract this, we warm-started each MLP from the previous DAgger iteration MLP.

However, this might increase the risk of being trapped in a local minima compounded by creating a sample imbalance in the dataset. To illustrate this, we consider the following numerical example.

Suppose each DAgger iteration rollout collects a dataset of size $N = 10000$

$\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N$. By iteration 9, the old dataset $\mathcal{D}_{\text{old}}$ (has 90,000 samples) heavily outweighs the newly added rollout data $\Delta\mathcal{D}$ (10000 samples). The contribution of the newly added data to the learning becomes increasingly diminished especially towards the latter DAgger iterations. This is particularly harmful because the newly collected samples become increasingly representative of the state distribution encountered by the learned policy in the later iterations where β approaches zero and the learner increasingly controls the system. Consequently, the policy may not sufficiently adapt to the newly encountered states, despite these being the most relevant for eventual deployment. While this might still happen even in the case of training the MLP from scratch, warm-starting from the last iteration MLP might further exacerbate this effect causing the MLP to become stuck in a local minimum. Utilizing the learning rate might not be suitable to mitigate this problem. By increasing it to escape the local minimum we risk destroying the features the MLP already learnt and then it becomes hard to fine tune. On the other hand, using a smaller learning rate means not being able to escape the local minimum. A dynamic learning rate can be used (higher in the beginning, decreases gradually), however, this was not needed since the upcoming point resolves the core issue.

Dataset Curation

The dataset size explodes towards the end, by the last couple of iterations the dataset size is more than 10x the original size. This problem became especially exacerbated during the generalization phase. The strategy is both inefficient because we need to train on too much data that is redundant and problematic due to the sample imbalance discussed earlier. A more efficient strategy is to train only on newly generated data or on a mix of new data and a representative sample of data from previous iterations.

The most valuable data for addressing covariate shift is collected in the final DAgger iterations, where $\beta \approx 0$ and the policy operates autonomously. These trajectories contain the states the learned policy actually visits, including realistic execution drift and recovery behaviors that are absent from the original expert demonstrations. As a result, they provide the strongest learning signal for the final policy.

In contrast, early iterations, where β is large (close to its maximum of 0.9), are dominated by expert intervention. The resulting samples are drawn primarily from $P(s \mid \pi_{\text{expert}})$ and are useful for initializing the policy, but they are less representative of the distribution encountered at deployment. Retaining all of this data throughout training therefore introduces substantial redundancy while contributing relatively little to the final behavior.

Reducing the dataset size offers two advantages: it lowers the computational cost of training and shifts the emphasis toward the distribution that matters most for autonomous control. A practical compromise is to retain all newly collected data from the final closed loop iterations while keeping only a representative subset of earlier data. For example, an 80% new and 20% sampled historical split preserves examples of recovery from rare, highly off nominal states without allowing redundant expert guided trajectories to dominate training.

The question then becomes how to choose the old data.

Possible Metrics to Curate the Data

Uniform: Uniform random subsampling of the retained old data will, by chance, mostly keep easy data points, since that makes up most of any episode, and only occasionally keep the rarer, difficult maneuver data points. This is not ideal, especially towards the later iterations. As more iterations are added, the difficult examples from earlier iterations are increasingly diluted by the much larger number of easy, near-duplicate samples, making it less likely that uniform sampling will retain the data that is actually most informative for correcting the policy.

Recency bias: Sample more heavily from iterations that are more recent, since they are more representative of the policy's current behavior. However, depending on recency alone walks into the same trap of representing mostly easy data points.

Hardness bias: If a data point is considered “easy,” e.g., the policy output and expert output are close, either dismiss it entirely or give it much less weight than “hard” points when sampling

the old data. This proved to be less straightforward than it initially seemed, since the difficulty of a data point can be influenced by the history of deviations and errors that preceded it. In some cases, the state itself was not particularly difficult, but because it occurred after a sequence of errors, it appeared to be a hard example according to our criterion. This caused the MLP to learn overly aggressive corrective behaviors, which ultimately degraded performance even further (see the Results section for more details).

Stratifying and binning: Guarantee diversity by classifying points based on maneuver location, such as loitering, turning, moving in a straight line, etc. This can be done using reference velocity and acceleration, and then taking a certain percentage from each bin. The percentage could, for example, be determined by a hardness score for each bin.

Multiple tests were conducted using these ideas and different combinations of them. The last two ideas, especially, were mixed in different ways. For example, the data points within each bin would be selected or not based on difficulty to ensure good representation of both the different maneuvers and their difficulty levels.

Final Dataset Management Strategy

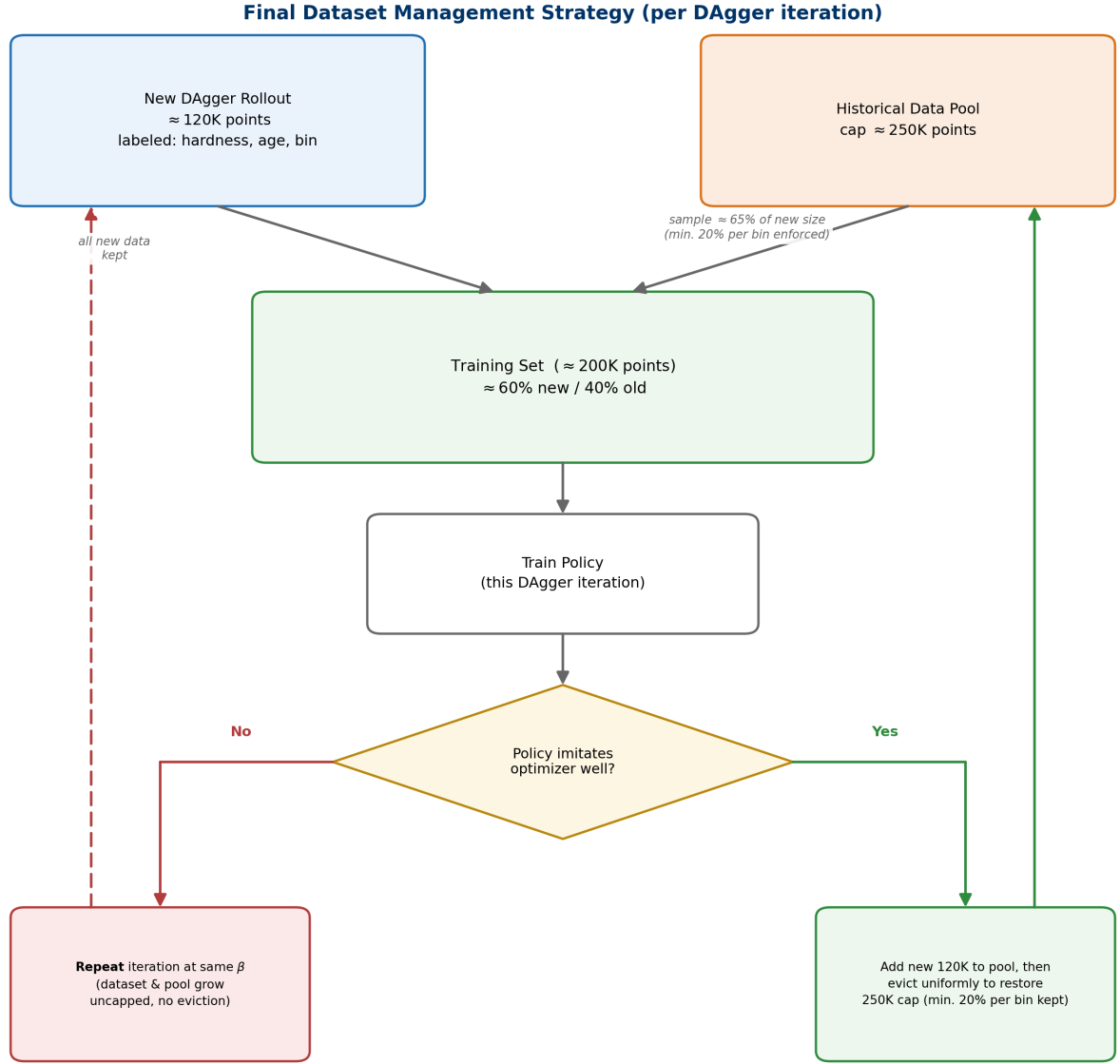
Each DAgger iteration generates approximately **120K new data points**, which we label with **hardness, age, and bin/category**. We keep all of this new data for training and additionally augment it with uniformly sampled old data from a historical pool. When sampling the old data, we enforce a minimum representation per bin. If a bin would contribute less than the minimum 20%, we sample additional points from that bin until this minimum is guaranteed.

The amount of old data sampled is around **65% of the size of the new data**, resulting in a final training set of approximately **200K points**, with roughly **60% new data and 40% old data**. After training is completed, the new 120K points are added to the old data pool. Since the pool has a fixed size of **250K**, some data is then uniformly sampled for eviction to bring the pool back to its target size, again respecting the same minimum percentage per bin.

If the evaluation of a DAgger iteration proves to be unsatisfactory, based on metrics measuring how well the policy imitates the optimizer, the iteration is repeated with the same β . During this process, both the training dataset and the data pool are allowed to expand without any sampling, meaning that all available data is used for training and all data is kept in the pool. Once the policy reaches satisfactory performance, we shrink the datasets back to their normal sizes and resume the standard sampling procedure.

The bins are defined as:

- **0 = Loiter:** reference speed $< V_{LO} = 0.05 \text{ m/s}$ (parked)
- **1 = Cruise:** not Loiter or Transition (steady motion)
- **2 = Transition:** reference $|\text{accel}| > A_{HI} = 0.02 \text{ m/s}^2$ (ramping)



Bins: **0** = Loiter (ref. speed < 0.05 m/s) **1** = Cruise (steady motion) **2** = Transition ($|\text{accel}| > 0.02 \text{ m/s}^2$)

Figure 4.2: Final dataset management strategy per DAgger iteration: new rollout data and a bin-stratified sample of the historical pool form the training set; after a successful iteration the new data is merged into the pool and excess points are evicted uniformly (respecting the per-bin minimum) to restore the size cap.

Evaluation metrics

Two main metrics are used to determine whether a DAgger iteration was successful and we can safely move to a lower β or not.

Jitter

For carrier i , we define jitter as

$$J_i = \frac{1}{T-1} \sum_{t=1}^{T-1} |\lambda_i(t) - \lambda_i(t-1)| \quad (4.8)$$

It is the mean absolute change between consecutive time steps of λ_i

- $J_i = 0$ means the carrier outputs a perfectly constant coefficient throughout the episode.
- Larger J_i means the policy changes its command more abruptly from one step to the next.
- There is no differentiation between increases and decreases from one step to the other since the absolute difference is used.

The team metric is simply the average across all drones:

$$J = \frac{1}{N} \sum_{i=1}^N J_i \quad (4.9)$$

Remark: this definition relies on our sampling period Δt which is 10ms in our implementation.

Although we refer to this metric as jitter, it actually measures the mean absolute first difference of the action sequence. Consequently, it captures overall temporal variation in the policy output, including both abrupt fluctuations and smooth, gradual changes. Therefore, the optimizer's smooth λ has value $J \sim 0.016$ and we consider a policy to be smooth enough if its $J < 0.02$.

Mean Squared Error

For drone i , the mean squared error is:

$$\text{MSE}_i = \frac{1}{T} \sum_{t=0}^{T-1} (\lambda_i(t) - \lambda_i^{\text{exp}}(t))^2 \quad (4.10)$$

Where λ_i is the policy's output and λ_i^{exp} is the optimizer's output.

The team-level MSE is the average across all drones:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \text{MSE}_i \quad (4.11)$$

An $\text{MSE} < 10^{-3}$ is considered satisfactory.

4.3.3 Generalization

Once the policy was successful on one trajectory and the basic methodology proved to be functioning, the next step was to generalize by training on multiple trajectories. Training was done by sampling random component-wise rest-to-rest quintic polynomial trajectories. A validation/testing set was held out during training to evaluate the policy's ability to generalize to unseen trajectories.

The generation of the random trajectories ensured generalization across movement scales, movement axes and episode durations. For example, testing was done on trajectories with displacements ranging from 0 to 20m along one or multiple axes. Furthermore, during training, episode durations were up to 35s, while the held-out test set included trajectories with durations of up to 75s. Thus, the evaluation included not only unseen combinations of movement magnitudes and axes, but also trajectories substantially longer than those encountered during training. This provides a test of whether the learned policy can maintain appropriate behavior over extended trajectories rather than relying on a fixed episode duration. Load orientations/rotations were limited to the range of $[-10^\circ, 10^\circ]$ for each Euler angle to reduce the risk of system destabilization.

4.4 Phase Two : Reinforcement Learning

4.4.1 Introduction

So far we have only discussed replicating the original optimizer. We can, in theory, deploy it in a separately on each carrier assuming perfect knowledge of the system by all of them. Since they will agree on everything and they employ the same algorithm, coordination will be achieved implicitly without the need for communication. Each carrier will calculate the full force $f(t) = f_g(t) + f_\lambda(t) \in \mathbb{R}^{12}$ (eq. (4.1)) and by knowing its ID, will apply its own slice from this vector $f_i(t) \in \mathbb{R}^3$. However, this property that might be helpful in theory to deploy the base controller in a decentralized fashion instead makes deploying it under realistic assumptions quite delicate.

The base controller uses global information on all the carriers to calculate $f(t)$. Specifically all the carrier velocities are needed to calculate λ . This global carrier information was reconstructed using load state and the total applied force $f(t)$. Once we have disagreements on the load state and the applied force (due to noise and delays/desynchronization), the reconstructed carrier velocities become completely corrupted and unreliable. This is on top of normal misalignments that will result from each carrier applying the controller to a different load state (from each carrier point of view).

While this misalignment is problematic, as long as the noise and desynchronization are bounded, the PID controllers will be able to stabilize the load. What is really insidious, is the

disagreement in λ due to the corruption of the global information. This estimation is delicate and therefore the other carrier states become practically inaccessible. This is catastrophic as even a slight disagreement in λ will continue to integrate over time and eventually lead to complete chaos and loss of coordination. To illustrate the issue, we consider the following example.

Suppose carrier 1 receives a slightly noisy and delayed version of the load state. Based on that, it calculates a slightly inaccurate force. While this force might not significantly affect the load, it might cause its own velocity to drop below the aerodynamic threshold, ϵ . Based on that, its optimizer/student policy decides to increase A and ξ . This is completely invisible to the other carriers, so their A and ξ remain constant. Over the next couple of seconds, a phase shift develops between the carriers' internal forces, and they begin to no longer cancel each other out, causing the load to swing. As time goes on, the disagreement becomes catastrophic, and each carrier's trajectory becomes completely uncoordinated with the others. This causes the carriers to crash into each other and the load to swing uncontrollably, as there is no feedback mechanism to make the carriers' internal forces coordinate again.

Our goal is to deploy the same controller, while training a residual policy that corrects/augments its action. The objective would be to find a policy that can still maintain reasonable load tracking and coordinated carrier trajectories despite:

- Noisy and asynchronous load state measurements
- Unreliable estimates of global information

Our setup is not simply attempting to use RL to produce better estimations of the global information; rather, it is investigating whether a learned policy can directly achieve robust decentralized behavior under the degraded information structure, without explicitly estimating or compensating for each agent's noisy and desynchronized observations.

In the results section, we argue that the policy achieves this goal by finding a control and coordination strategy that works under incomplete and noisy information, not by trying to "correct and denoise" this information.

Overall Architecture

Since we have multiple agents operating independently but all affecting the load together and affecting each other, the natural choice is to frame this problem as a Multi-Agent Reinforcement Learning (MARL) problem. Additionally, the performance of the carriers will be evaluated using a combination of global states (such as the true load position) and local states (such as the carrier's own velocity). Therefore, a Centralized Training with Decentralized Execution (CTDE) architecture was selected. This means that access to the global true centralized information is only available during the training evaluation process and not available to the individual carriers during the execution. We achieve this by using a shared/centralized critic with access to global state while the actor is deployed decentralized on each carrier. This means that

during every time step, we have four actors taking actions in parallel and a central critic evaluating the system state. Since the carriers are identical and perform the same control task, differing only in their observations, we chose to share the same actor policy across them. This means that the four actors are homogeneous, using the same policy network and parameters, while producing actions based on their respective observations. The network architecture is a 4-layer MLP of size [98, 256, 256, 10] for the actor's policy and a 4-layer MLP of size [76, 256, 256, 1] for the centralized critic.

Furthermore, we use Multi-Agent Proximal Policy Optimization (MAPPO), akin to another work [8] that tackled a similar, yet distinct, decentralized multi-carrier cooperative payload transportation problem. While in [8] the carriers were multirotors that could hover, this work tackles non-stopping (fixed-wing) carriers. The MAPPO algorithm also aligns naturally with our chosen CTDE architecture.

4.4.2 Action Space

Originally, the idea was to keep wrench controller and optimizer/student policy and just add a residual δf force to correct their behavior.

$$f(t) = f_g(t) + f_\lambda(t) + \underline{\delta f(t)} \quad (4.12)$$

However, this means that the policy will be forced to learn the underlying force decomposition if it wants to act specifically on the internal forces for example. Instead of having the policy search until it discovers this division, it is more fruitful to make the action space work directly on both the wrench $w(t)$ and lambda $\lambda(t)$.

$$f(t) = G(t)^\dagger \cdot (w_d(t) + \underline{\delta w(t)}) + N(t) \cdot (\lambda(t) + \underline{\delta \lambda(t)}) \quad (4.13)$$

where $\delta w \in \mathbb{R}^6$ and $\delta \lambda \in \mathbb{R}^4$, total action space $\in \mathbb{R}^{10}$

This makes the policy gain insight into the underlying mechanics of the problem without having to discover it from scratch, as gradients on both δw and $\delta \lambda$ are separate therefore learning how each one of them affects the system is more straightforward.

4.4.3 Observation Space

The actor observation space must be limited to only local information (i.e. data that can be measured locally by sensors onboard the carrier). Additionally, in this work we assume no mid-flight communication, so the agent cannot have information about other carriers (unless they are mathematically reconstructed from local observations).

Actor's Observation Space (Original³) $\in \mathbb{R}^{44}$

1. Measured load state $\in \mathbb{R}^{18}$:

$$[p_L(t), R_L(t), \dot{p}_L(t), \omega_L(t)]$$

Naturally the most important observation is the load state.

Since there is no direct communication between the carriers, the load state is the only implicit way of information sharing between them. The underlying assumption is that the policy learns to use it to partially infer some information about the other carriers and consequently what its own action should be.

$\hookrightarrow$ **Noise and desynchronization**

To fully test the decentralization, we assume that every carrier makes its own measurement that is affected by its own noise and delay/desynchronization.

The noise was modelled as an autoregressive Gaussian noise of order one AR(1). It represents temporally correlated estimation error. By tuning the correlation coefficient, it can represent a drifting, bias-like, viewpoint-dependent pose-estimation error. It was shown that this type of noise is more representative of realistic sensor noise rather than independent Gaussian white noise [12].

Additionally, each measurement is randomly delayed by $d_i \in [0, 2]$ time steps. Since our sampling period is 10 ms, this means that each measurement is delayed between 0 and 20 ms. d_i is randomly chosen initially and evolves according to a random walk. This is done to represent asynchronous measurements by different carriers. The selected values are not derived from a specific empirical study; instead, they were chosen as reasonable estimates within the expected order of magnitude. A detailed characterization of these parameters is beyond the scope of this work since the characteristics of measurement delays and sensor noise can vary considerably depending on specific sensor and measurement setup [13].

2. Own carrier position and velocity $\in \mathbb{R}^6$:

$$[p_R(t), \dot{p}_R(t)]$$

This observation is needed because we want to enable the policy to learn the orbiting behavior. Therefore, knowing the current position and velocity vector is necessary to determine the next action to maintain the orbiting behavior.

³The use of the term “original” is clarified later in this section.

3. Base force components $\in \mathbb{R}^6$:

$$[f_g(t), f_\lambda(t)]$$

These are the forces the policy is supposed to augment/correct. Because the policy is a residual one, it needs to know what the base controller is outputting in order to decide its appropriate action. The reason the force components are split is to provide specific information on what the base controller is doing both to the payload forces and to the internal forces. This also mirrors the action space structure.

4. Fourier time embedding/features $\in \mathbb{R}^{14}$:

$$\phi(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \sin(\omega_2 t), \cos(\omega_2 t), \dots, \sin(\omega_7 t), \cos(\omega_7 t)]$$

Similar to the observation space of the IL policy, the Fourier time embedding is provided to the policy to anchor it in time, allowing it to learn the periodicity of the orbiting behavior without being able to explicitly memorize specific actions at specific times. Similar to the IL policy, we use seven frequencies $\{1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0\}$ rad/s based on the same reasoning.

To summarize, the original observation space $\in \mathbb{R}^{44}$ and consists of:

1. Measured load state $\in \mathbb{R}^{18}$

$$[p_L(t), R_L(t), \dot{p}_L(t), \omega_L(t)]$$

2. Own carrier position and velocity $\in \mathbb{R}^6$

$$[p_{R_i}(t), \dot{p}_{R_i}(t)]$$

3. Base force components $\in \mathbb{R}^6$

$$[f_g(t), f_\lambda(t)]$$

4. Fourier time embedding/features

$$\phi(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \sin(\omega_2 t), \cos(\omega_2 t), \dots, \sin(\omega_K t), \cos(\omega_K t)]$$

Over the duration of this work, multiple experiments with different observation spaces were conducted. The goal of these experiments was not only to assess whether better performance can be achieved with different observation spaces, but also to shed some light on how the policy might be functioning.

For example, historical measurements were added for the load state and the carrier's own position and velocity. However, this did not lead to an improvement in performance. The lack of

benefit from additional historical information hints that the policy is not trying to estimate “more correct” global information. Instead, this suggests that the policy is finding control actions that work despite loss of information. Nevertheless, we include the augmented observation space as it was used to conduct many other experiments in the latter part of the work, on the possibility that it might lead to some minor improvement.

Actor's Observation Space (Augmented) $\in \mathbb{R}^{98}$

1. Original Observation Space in addition to:
2. The last two measured load states $\in \mathbb{R}^{24}$

$$[p_L(t-1), R_L(t-1), \dot{p}_L(t-1), \omega_L(t-1)]$$

$$[p_L(t-2), R_L(t-2), \dot{p}_L(t-2), \omega_L(t-2)]$$

3. The last two own carrier position and velocity $\in \mathbb{R}^{12}$

$$[p_R(t-1), \dot{p}_R(t-1)]$$

$$[p_R(t-2), \dot{p}_R(t-2)]$$

4. Reference load state at current time $\in \mathbb{R}^{18}$

$$[p_{Ld}(t), R_{Ld}(t), \dot{p}_{Ld}(t), \omega_{Ld}(t)]$$

As mentioned earlier, the goal of adding historical measurements of the load state and the carrier's own position and velocity was to determine whether the policy can benefit from estimating better information about the system. Furthermore, the reference load state was added to give the policy insight about the base controller's action and the possible direction of evolution of the load state in the next time step. We will show in the results section that these additions did not lead to any meaningful enhancement of the performance and we will comment further on what they mean.

Critic state $\in \mathbb{R}^{76}$

Unlike the actor's observation space, the critic's state has access to global information that is true (not noisy or delayed). This allows it to objectively evaluate the actors' actions and provide useful gradients for them to improve.

1. True load state $\in \mathbb{R}^{18}$:

$$[p_L(t), R_L(t), \dot{p}_L(t), \omega_L(t)]$$

Used to assess whether the load is following its reference trajectory or not.

2. All carriers' positions $\in \mathbb{R}^{12}$:

$$[p_{R_1}(t), p_{R_2}(t), p_{R_3}(t), p_{R_4}(t)]$$

3. All carriers' velocities $\in \mathbb{R}^{12}$:

$$[\dot{p}_{R_1}(t), \dot{p}_{R_2}(t), \dot{p}_{R_3}(t), \dot{p}_{R_4}(t)]$$

Carrier positions and velocities are used to determine if they are performing the orbiting behavior correctly or not.

4. Expert-generated carriers' positions $\in \mathbb{R}^{12}$:

$$[p_{R_{e1}}(t), p_{R_{e2}}(t), p_{R_{e3}}(t), p_{R_{e4}}(t)]$$

Based on each carrier's deviation from its respective expert trajectory, the critic should be able to evaluate how much degradation in the coordination between the carriers have occurred. The difference between the carriers' actual state and expert calculated one generally is the most affected/degraded metric when loss of coordination occurs and it is usually a good indicator on the overall health of the system. While the carriers are not required to follow the expert trajectory exactly, it is still useful as a qualitative proxy for smooth orbiting. The expert generated trajectories are calculated based on a centralized controller + optimizer that get clean global information. For each training trajectory we calculate and store the expert trajectories for the carriers, they are then fed to the critic. The goal is to learn the underlying orbiting behavior in general, hoping that it will generalize this behavior to new trajectories. Since this information is only available to the critic during training, we do not need to worry about having to generate them at deployment.

5. Delay of each carrier load measurement $\in \mathbb{R}^4$:

$$[d_1, d_2, d_3, d_4]$$

Delays are added to the critic state since they can make the state harder or easier, thus changing the expected reward, giving it to the critic helps making the advantage function more accurate. In other words, identical local observations by the carriers could correspond to different underlying difficulty based on how much desynchronization is between them. Therefore, to judge their actions and the resulting state of the system it is important for the critic to understand how delayed was each carrier, and how delayed they are with respect to each other.

6. Reference load state at current time $\in \mathbb{R}^{18}$

$$[p_{Ld}(t), R_{Ld}(t), \dot{p}_{Ld}(t), \omega_{Ld}(t)]$$

Initially the reference load state was not provided to the critic. The rationale was that it will be able to infer it from the expert-generated carriers' positions. While this proved to be true in our setup, it is not in general true, as the load has extra degrees of freedom based on the cables' directions. However, in our setup with clean states, the expert is able to drive the load with practically zero swinging, making the reference load state geometrically locked to the carriers' state. In practice, the critic was able to learn this relationship quite well, as it was able to achieve excellent levels of accuracy (very low loss value), confirming our reasoning. Nevertheless, as a matter of principle, we still added it in the latter part of the work as a precaution, since the assumption might be violated at some point.

To summarize, the critic state $\in \mathbb{R}^{76}$ is composed of:

1. True load state $\in \mathbb{R}^{18}$:

$$[p_L(t), R_L(t), \dot{p}_L(t), \omega_L(t)]$$

2. All carriers' positions $\in \mathbb{R}^{12}$:

$$[p_{R_1}(t), p_{R_2}(t), p_{R_3}(t), p_{R_4}(t)]$$

3. All carriers' velocities $\in \mathbb{R}^{12}$:

$$[\dot{p}_{R_1}(t), \dot{p}_{R_2}(t), \dot{p}_{R_3}(t), \dot{p}_{R_4}(t)]$$

4. Expert-generated carriers' positions $\in \mathbb{R}^{12}$:

$$[p_{R_{e1}}(t), p_{R_{e2}}(t), p_{R_{e3}}(t), p_{R_{e4}}(t)]$$

5. Delay of each carrier load measurement $\in \mathbb{R}^4$:

$$[d_1, d_2, d_3, d_4]$$

6. Reference load state at current time $\in \mathbb{R}^{18}$

$$[p_{Ld}(t), R_{Ld}(t), \dot{p}_{Ld}(t), \omega_{Ld}(t)]$$

4.4.4 Reward function

The design of an efficient reward function is generally one of the most intricate challenges in RL. Fortunately, in the case of residual RL the challenge is slightly simplified since we already have a starting point. In this work, the design methodology was to observe failure modes of the existing setup and add terms one by one to steer the behavior away from these failure modes

and towards desirable behavior.

The per drone reward function is:

$$r_i = - (w_{\text{manifold}} d_i^2 + w_{\text{stall}} \text{stall}_i + w_{\text{load}} \|e_L\|^2 + w_{\text{overspeed}} \text{overspeed}_i^2 + w_{\text{overshoot}} \text{overshoot}_i) \quad (4.14)$$

where w_{manifold} , w_{stall} , w_{load} , $w_{\text{overspeed}}$ and $w_{\text{overshoot}}$ are all strictly positive weights

And the team reward simply is:

$$R_{\text{team}} = \sum_{i=1}^N r_i \quad (4.15)$$

Additionally, a penalty of -100 is applied once if any carrier's speed exceeds $v_{\text{blowup}} = 100$ m/s. This also leads to the truncation of the episode.

In this subsection we break down each term one by one:

- **manifold (d_i^2):**

$$d_i^2 = \|p_{\text{expert},i}(t) - p_i(t)\|^2 \quad (4.16)$$

Where $p_{\text{expert},i}(t)$ is the expert-generated trajectory for carrier i at time t (centralized controller with true states) and $p_i(t)$ is the actual carrier position. Naturally, this is a per carrier term.

If there is no noise or desynchronization between the drones the value of this term should be zero. This term was added as a result of experimentation, so it is not required for the policy to follow the expert-generated trajectory exactly. However, it was discovered that it provides a useful feedback on the general health of the system. Usually it is low if the carriers are orbiting correctly and it is high if they are flying around uncontrollably or potentially crashing into each other. The squared absolute difference is used for this purpose, as it does not punish small magnitude deviations (minimizes their influence) while placing greater emphasis on bigger deviations.

- **Aerodynamic stall (stall_i):**

$$\text{stall}_i = d^2 + w_{\text{lin_stall}} \cdot d \quad (4.17)$$

where $d = \max(0, (\varepsilon + \text{margin}) - \|v_i\|)$. ε is the minimum flight speed before stall, margin is a small positive number to disincentivize approaching ε , $\|v_i\|$ is the velocity norm of carrier

i , and $w_{\text{lin_stall}}$ is a strictly positive weight

Again, this is a per carrier term. This term is a central reward term: no matter what the carriers do, they should never go below the aerodynamic stall speed ε . Therefore, this term was included from the outset and was not introduced as a result of experimentation. While some other terms are included to help shape the gradient and guide the policy toward good solutions, they are not inherently part of the requirements. This term, however, is a non-negotiable requirement. Therefore, this is reflected in the selection of its weights, which are chosen to be relatively high. Additionally, a margin was added to provide a buffer before reaching the actual stall velocity. Furthermore, linear and quadratic difference terms are used together to penalize violations immediately, even when they are small, while also greatly penalizing larger violations. The linear term is particularly important for small violations, for which the quadratic term alone would not provide a sufficiently strong penalty, while the quadratic term places greater emphasis on deep stalls.

- **load tracking error ($\|e_L\|^2$):**

$$e_L = p_{Ld}(t) - p_L(t) \quad (4.18)$$

Similar to the aerodynamic stall term, this term was included from the beginning as it is a main requirement of the system. Nevertheless, unlike the aerodynamic stall term, there is some leeway if the magnitude of the error is small. Accordingly, only quadratic error is used and no linear penalty is applied. This means that small magnitude errors are minimized while larger ones are emphasized. Moreover, this term is the only term that is shared between all the carriers.

- **Carrier Overspeed (overspeed_i^2):**

$$\text{overspeed}_i^2 = \min(\text{over}^2 + w_{\text{overspeed_linear}} \cdot \text{over}, \text{cap}_{\text{overspeed}}) \quad (4.19)$$

where $w_{\text{overspeed_linear}}$ is a positive weight, $\text{over} = \max(0.0, \|v_i\| - \text{overspeed}_v)$, $\|v_i\|$ is the velocity norm of carrier i , and overspeed_v is the upper bound of acceptable velocity norms and $\text{cap}_{\text{overspeed}}$ is an upper bound on the whole term value

This term was added by observing how the base controller fails on many trajectories. In general, it was observed that a lack of coordination between the carriers can cause their speeds to increase dramatically, ultimately resulting in system blow-up, with speeds reaching unrealistic values (hundreds of m/s or more). We used the same combination of linear and quadratic penalties as before to discourage both small and large violations. Additionally, this term was capped to an upper value to prevent the advantage values, returns and ultimately the PPO update from becoming numerically unstable. This is because, as mentioned earlier, the carriers' velocities can sometimes spike to arbitrary large values. This penalty is applied per carrier to only penalize the carrier that is accelerating.

- $\delta\lambda$ **cap overshoot** (**overshoot_i**):

$$\text{overshoot}_i = \min(\text{over}_{\lambda,i}^2 + \text{over}_{\lambda,i}, \text{cap}_{\text{overshoot}}) \quad (4.20)$$

where $\text{cap}_{\text{overshoot}}$ is an upper bound on the whole term value, $\text{over}_{\lambda,i} = \max(0, \text{sat}_{\lambda,i} - 1)$, $\text{sat}_{\lambda,i} = \|\delta\lambda_i\|_2 / (\kappa_\lambda \cdot \|\lambda_{i,\text{base}}\|_2)$, and κ_λ is the allowable fraction of the base controller magnitude.

In order to prevent the residual policy from taking arbitrarily large actions, $\delta\lambda$ and δw were capped to a fraction κ_λ of the base controller values. This prevents the policy from driving the system to instability while still allowing it to explore all the possible action space without being penalized for it. However, during the training procedure, as the policy gets exposed to less-explored trajectories, it can start by outputting actions that are higher than these bounds. This causes a problem of dead gradients; decreasing or increasing the action slightly doesn't lead to any change in the system's behavior so the policy is not able to learn. Consequently, the goal of this penalty is to discourage the policy from outputting values that exceed the bound, even if such values were reached inadvertently. This problem was only observed for $\delta\lambda$ so the penalty is only applied there. δw always stayed below its upper bound.

The following is a table summarizing each term:

Term	Expression	Role
manifold (d_i^2)	$\ p_{\text{expert},i}(t) - p_i(t)\ ^2$	Stay on the expert loiter loop (generated from ground truth states). The goal is not to follow it exactly, but to learn the underlying orbiting behavior in general.
stall_i	Graced hinge below cruise floor: $d^2 + w_{\text{lin_stall}} \cdot d$, where $d = \max(0, (\varepsilon + \text{margin}) - \ v_i\)$	Fixed-wing stall floor (prevents airspeed drop).
load tracking error	$\ e_L\ ^2$ where $e_L = p_{Ld}(t) - p_L(t)$	Keep the load on reference
overspeed_i²	$\min(\text{over}^2 + 2 \cdot \text{over}, 75)$ where $\text{over} = \max(0.0, \ v_i\ - \text{overspeed}_v)$	Prevent carrier from speeding up unnecessarily. Blowup guardrail.

overshoot_i	$\min(\text{over}_{\lambda,i}^2 + \text{over}_{\lambda,i}, 9)$ where $\text{over}_{\lambda,i} = \max(0, \text{sat}_{\lambda,i} - 1)$ and $\text{sat}_{\lambda,i} = \ \delta\lambda_i\ _2 / (\kappa_\lambda \cdot \ \lambda_{i,\text{base}}\ _2)$ κ_λ : the allowable fraction of the base controller magnitude	Prevents the policy from taking actions above the allowed fraction of the base controller magnitude (dead gradient, no learning). The cap prevents the policy from taking overly aggressive actions.
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Truncation Condition: A one-shot penalty of -100 is triggered on a guard-truncated step if speed exceeds $v_{\text{blowup}} = 100$.

4.4.5 Centralized Controller Policy Guidance experiment (using Supervised Learning)

To test whether the policy is learning to recover the centralized controller behavior or genuinely learning a different decentralized reactive policy, we conducted an experiment in which we warm-started the policy network from a network that was trained using supervised learning to output $\delta\lambda^*$ and δw^* . Where $\delta\lambda^* = \lambda_{\text{base}} - \lambda_{\text{expert}}$ and $\delta w^* = w_{\text{base}} - w_{\text{expert}}$. λ_{base} and w_{base} are the base controller output when it is deployed decentralized on noisy and asynchronous measurements, while λ_{expert} and w_{expert} are the centralized controller output using clean (no noise and desync) states and global information. Furthermore, during training, the supervised learning loss term was added to the PPO loss terms. This provides a consistent gradient signal to the policy, encouraging it to learn solutions that recover the behavior of the centralized controller. However, as will be discussed in the results section, this approach was not effective and was therefore abandoned.

4.4.6 MAPPO implementation details

In this section we go over some design choices that we consider noteworthy. We do not go over all the MAPPO algorithm details as this is not the main focus of this work.

Advantage Estimates

For each of the four agents, a separate advantage estimate is computed using its individual reward and the shared global value estimate:

$$A_t^i = \text{GAE}(r_t^i, V(s_t)). \quad (4.21)$$

Thus, the critic provides a common assessment of the team's expected future performance from the complete global state, while the individual reward determines how the contribution of

each agent is reflected in its corresponding advantage. This allows the four agents to share a common global assessment while retaining distinct learning signals based on their individual contributions.

Critic Target

The target used to train the critic is computed as the mean of the individual agent rewards, together with the value estimate of the subsequent state:

$$y_t = \frac{1}{4} \sum_{i=1}^4 r_t^i + V(s_{t+1}). \quad (4.22)$$

This provides the critic with a target that reflects the collective performance of the four agents while accounting for the expected future return.

Curriculum training

Training the policy on the full set of trajectories simultaneously, with all reward terms active, proved challenging due to the diversity of behaviors and operating conditions. To facilitate policy learning, a curriculum training strategy was therefore adopted, in which both the training trajectories and reward terms were progressively expanded during training. This allows the policy to first learn from a more limited set of trajectories with simpler objectives before being exposed to increasingly diverse scenarios and more complex objectives.

Training on multiple trajectories together proved to be especially difficult, as the reward becomes extremely high variance, making the learning incredibly slow. This was not an issue of observation aliasing since the load reference trajectory is provided in the observation space. It is also not a network capacity since multiple attempts to increase the size of the MLP were made and the problem persisted. To combat this high variance, the policy was first trained on a single trajectory until mastery, and then more trajectories were added one by one. However, this still proved to be challenging and does not force the policy to learn to generalize. To ensure the correct diagnosis of the problem, and to explore a potential solution, we conducted the experiment illustrated in the next subsection.

Behavior Cloning (BC) of multiple policies

In order to test whether the policy is unable to learn multiple trajectories together due to observation aliasing or network capacity, multiple teacher policies $\{\pi_1, \dots, \pi_K\}$ were first trained separately on different trajectories until mastery. After that, their combined input-output pairs were used to train a single student policy/network of the same architecture using BC.

The combined dataset is:

$$\mathcal{D} = \bigcup_{i=1}^K \{(o_t, \pi_i(o_t))\} \quad (4.23)$$

where o_t is the observation and $\pi_i(o_t)$ is the teacher policy action

The single student policy π_θ with the same architecture and capacity is trained on $\mathcal{D}$ by minimizing the supervised imitation loss

$$\mathcal{L}_{\text{BC}}(\theta) = \mathbb{E}_{(o,a) \sim \mathcal{D}} [-\log \pi_\theta(a \mid o)] . \quad (4.24)$$

If the student policy is able to mimic the combined behavior of all teacher policies, this suggests that the main challenge is learning the policy, rather than the problem being inherently ill-posed or requiring a larger or more complex network. Additionally, since the combined student policy was trained using supervised learning in open loop, it might not be able to achieve the same performance as the teacher policies when deployed in closed loop. Instead of resorting to DAgger or other IL methods, the combined student policy can be used to warm-start a policy to be fine-tuned with RL to improve its behavior across the full set of trajectories mastered by the teacher policies. This process can later be iteratively repeated by combining the resulting policy with other policies that were trained in the same way, until generalization is achieved.

Overview of MAPPO Training Loop and Environment Step

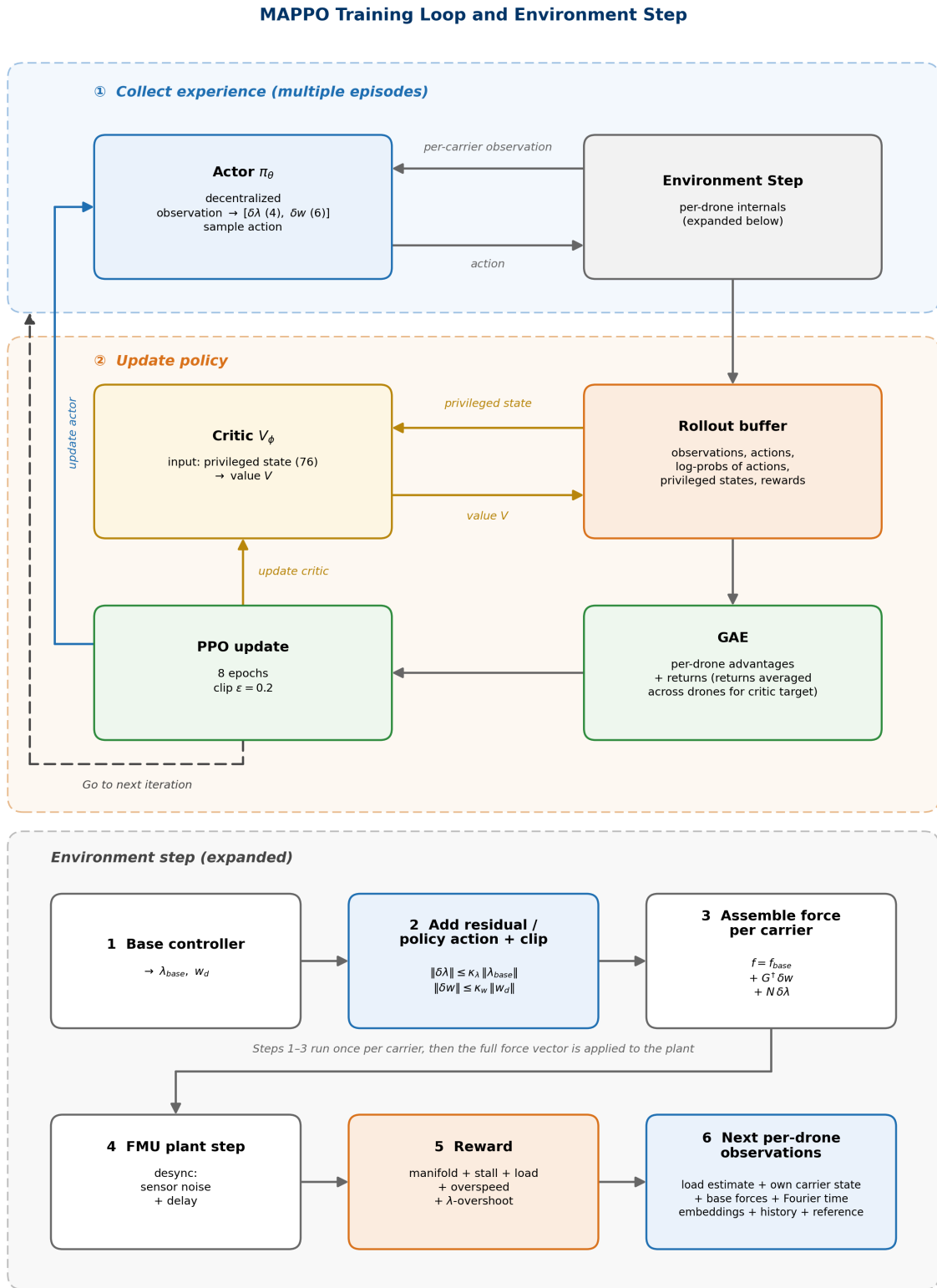


Figure 4.3: Overview of the MAPPO training loop and the environment step. Experience is collected by the decentralized actors interacting with the environment and stored in the rollout buffer. The critic evaluates the privileged states to produce the values used by GAE, and the resulting advantages and returns are used by the PPO update to update both the actor and the critic. The environment step is expanded in the bottom panel.

CHAPTER 5

Results and Discussion

5.1 Phase One : Imitation Learning

5.1.1 Single trajectory

Initially, the policy was trained on a single trajectory. The goal was just to test whether it will be able to learn to overfit this single trajectory or no. If it is not able to, then there is no reason to attempt to generalize and the observation space, network architecture ...etc needs to be revised. If it is able to learn the single trajectory, then we can proceed with the generalization and train on multiple trajectories. Many solutions were tested in this phase until a working one was reached (that was discussed in the Methodology chapter).

Reference trajectory

The load reference trajectory, i.e. the goal of the whole system, is a straight move of 11 m along the x -axis over 25 s, with no motion along the y and z axes. The load is static at the beginning and at the end, and moves at a constant velocity of 1.1 m/s for 10 s in between. This trajectory was chosen to engage the optimizer and force it to change A and ξ mid-trajectory.

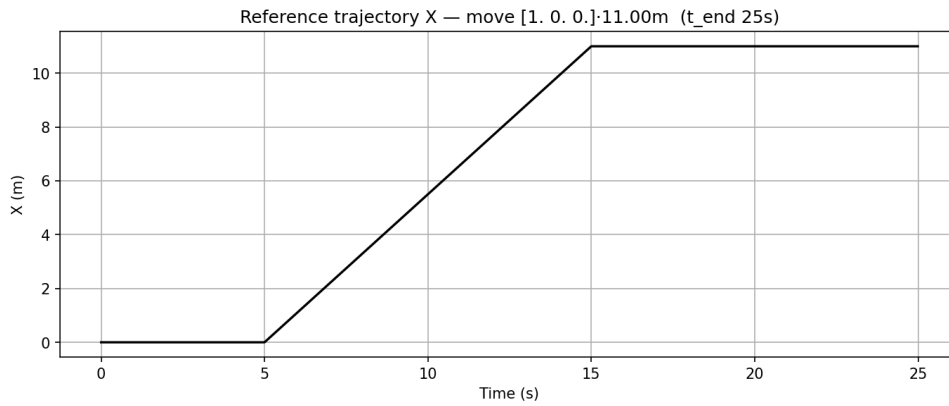


Figure 5.1: Load reference trajectory along the x -axis ($t_{end} = 25$ s).

Expert behavior

The figures below show the Classical Control Stack (CCS - the expert using the original optimizer) behavior on the reference trajectory: a 11 m move along the x -axis over 25 s. This

behavior serves as the reference that the proposed imitation learning (IL) policy aims to imitate. These figures were obtained by implementing the methodology presented in [5] and are shown here to provide the reader with a reference for the behavior being imitated.

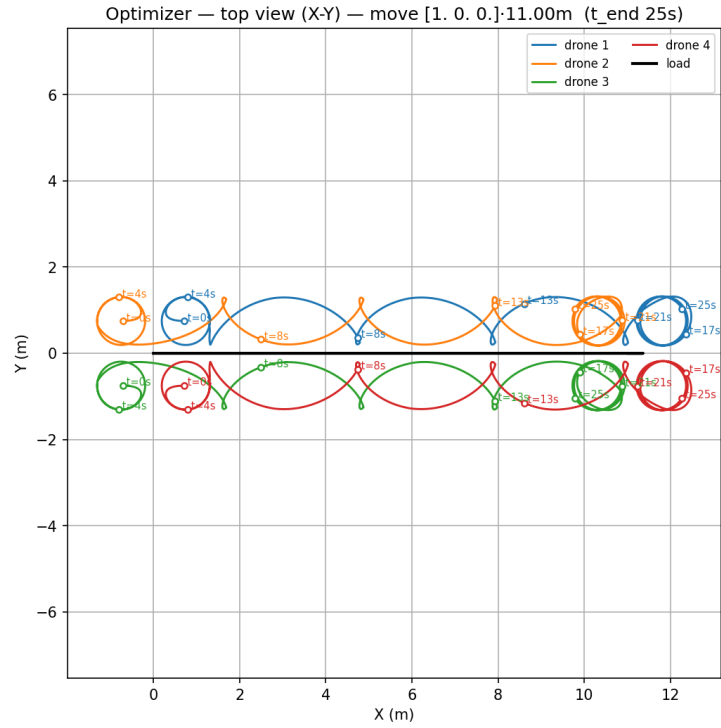


Figure 5.2: CCS carrier and load trajectories top view (X-Y) for a move of 11 m along x ($t_{end} = 25$ s).

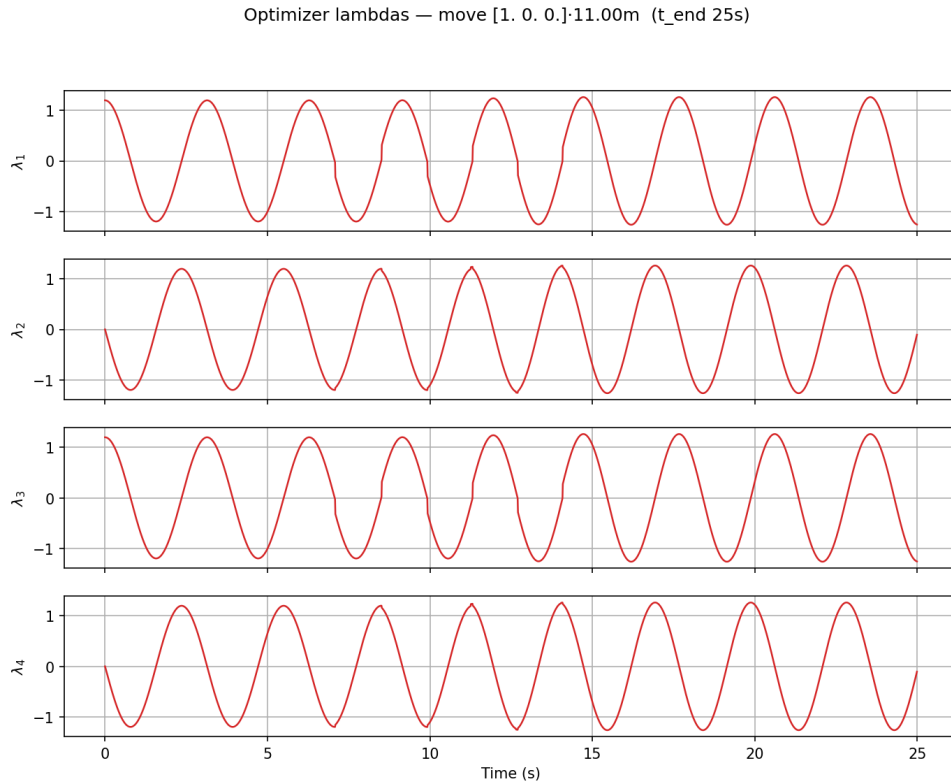


Figure 5.3: Optimizer λ values for the same trajectory.

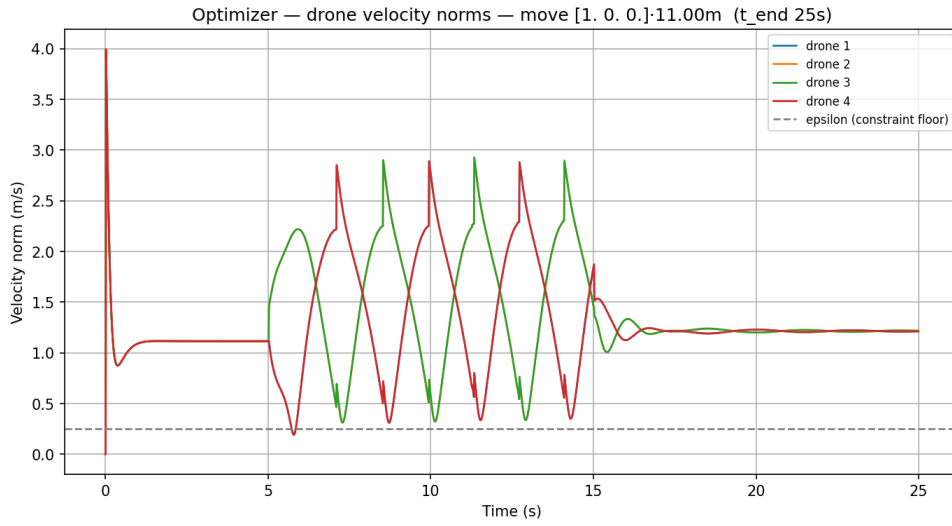


Figure 5.4: CCS carrier velocity norms for the same trajectory.

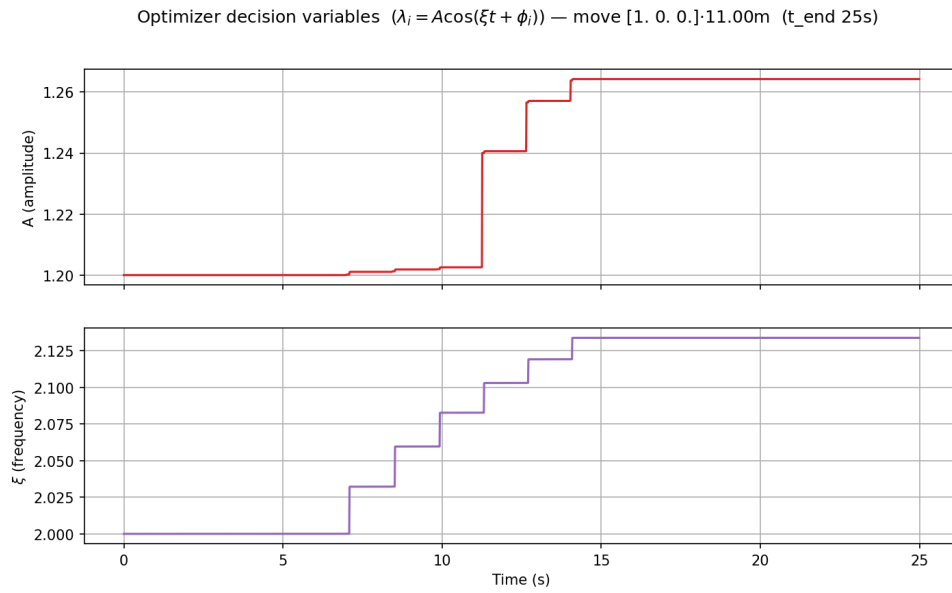


Figure 5.5: Optimizer decision variables $\lambda_i = A \cos(\xi t + \phi_i)$ for the same trajectory.

We see in the two figures above that as soon as the minimum velocity is violated the optimizer reacts and changes A and ξ and accordingly the velocity norms change and do not dip below epsilon again.

5.1.1.1 Supervised Learning

Initially, the policy was trained with supervised learning only. In this section, we show how it performs and why it was not enough.

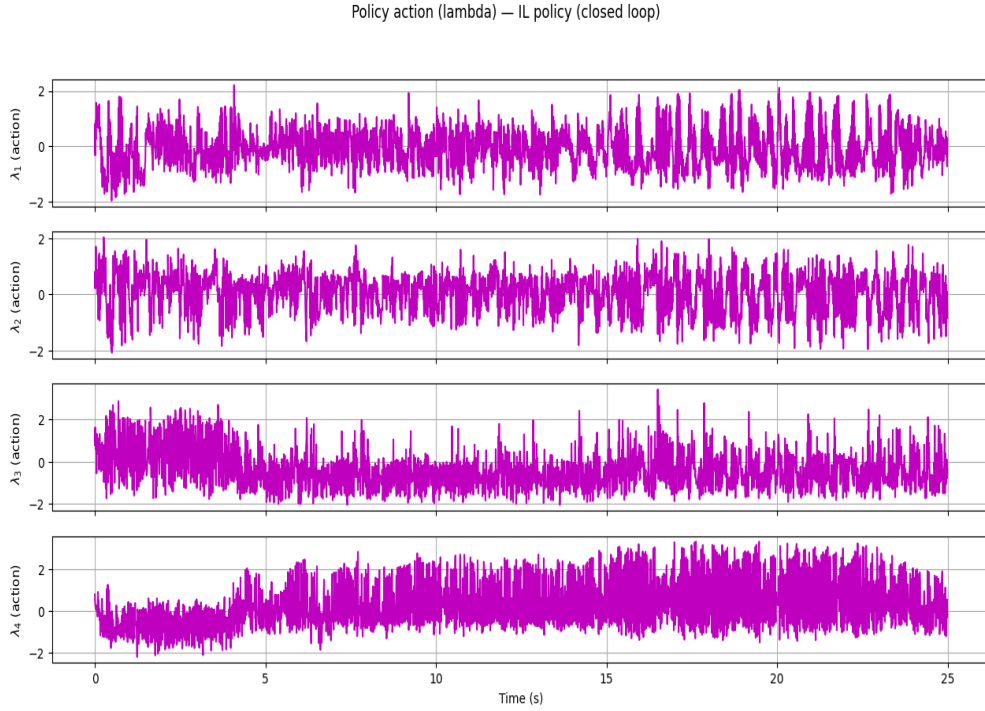


Figure 5.6: λ produced by the supervised-only policy while flying, showing uncontrolled jitter.

Even though the MSE after the supervised learning was $< 1 \times 10^{-3}$, deploying the policy in closed loop is much more unstable and leads to huge jittering in λ due to the distribution shift discussed in Section 4.3.2. For this reason, using supervised learning alone is not enough and DAGger becomes necessary.

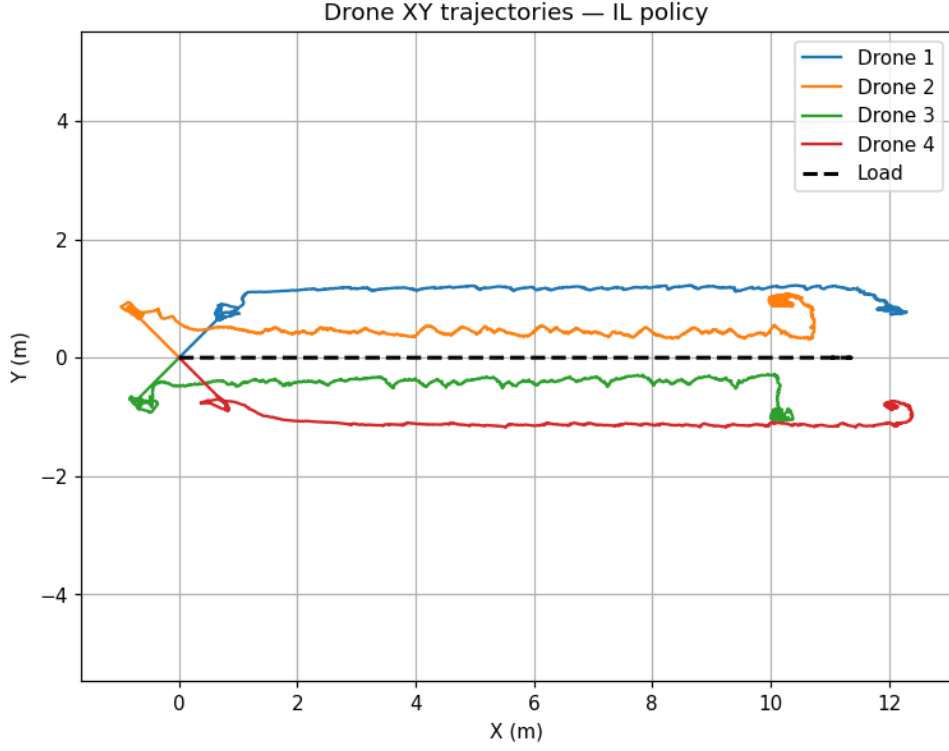


Figure 5.7: Carriers and load Trajectory top view (X-Y) trajectories of the policy trained with supervised learning only.

We notice that the load trajectory is not changed (still following the reference) even though the λ 's are completely corrupted. This is due to the strict separation of the internal forces (that operate in the nullspace of the grasp matrix) and the forces affecting the load. The internal forces only affect the carrier trajectory partially, however, it is enough to make it physically unfeasible. These results must be interpreted with care, as the simulation might indicate that the load tracking is working or that the carriers are following a jittery trajectory, while if we run this in real life it will probably fail due to the high jitter.

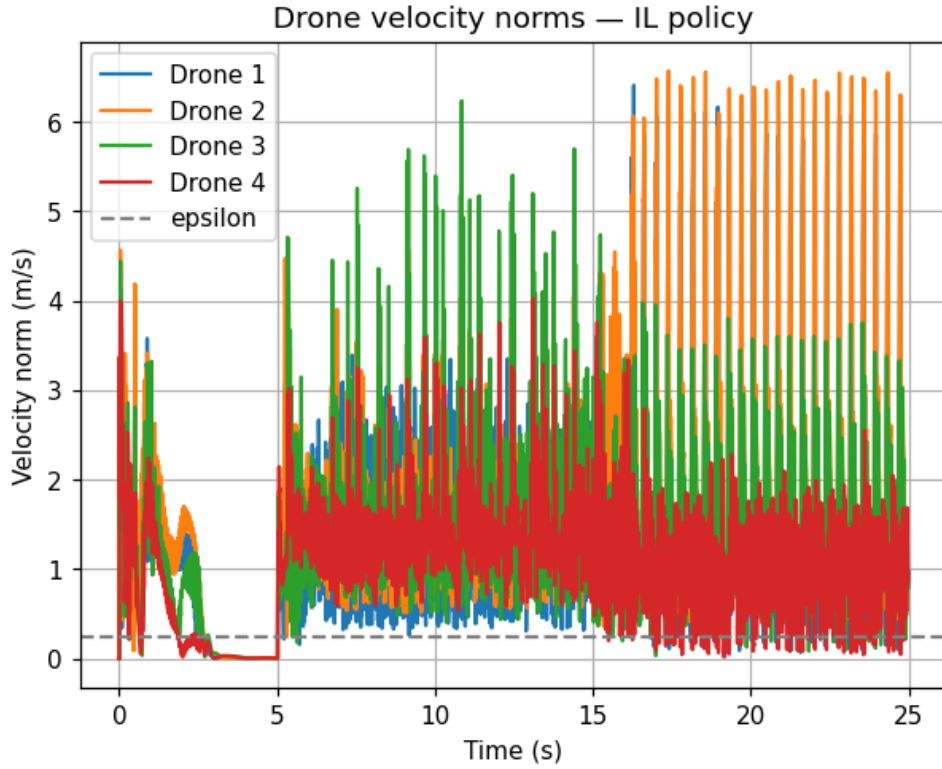


Figure 5.8: Drone velocity norms of the policy trained with supervised learning only.

This plot again highlights the same idea mentioned before. The load tracking its reference trajectory in the simulation does not mean that it will actually do in real life. This plot shows that the carriers' velocities are very jittery and dip many times below epsilon and even reach zero for a considerable period of time. Therefore we can firmly conclude that the policy is failing. A plant model that captures these interactions can better reflect what will happen in real life but would be much more complicated to build and is outside the scope of this work. Until this more accurate plant model is used, all resulting plots must be evaluated with care to determine if the policy is successful or no.

5.1.1.2 DAgger

The following figures compare the DAgger-trained policy (network) against the optimizer (CCS) on the single reference trajectory.

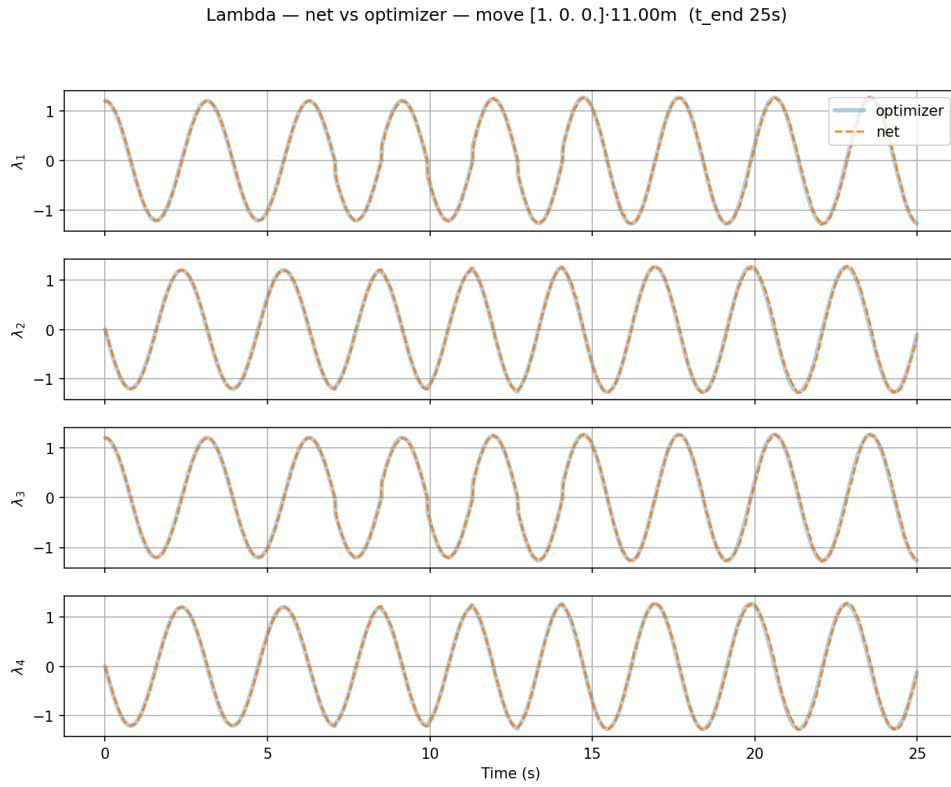


Figure 5.9: λ per carrier: DAgger-trained policy (network) vs optimizer.

As it can be seen in this plot, the DAgger-trained policy produces near-identical output to the optimizer. This is the main output and everything else is downstream from it. Therefore, we expect the other plots to show satisfactory performance. Refer to table 5.1 for a numerical comparison.

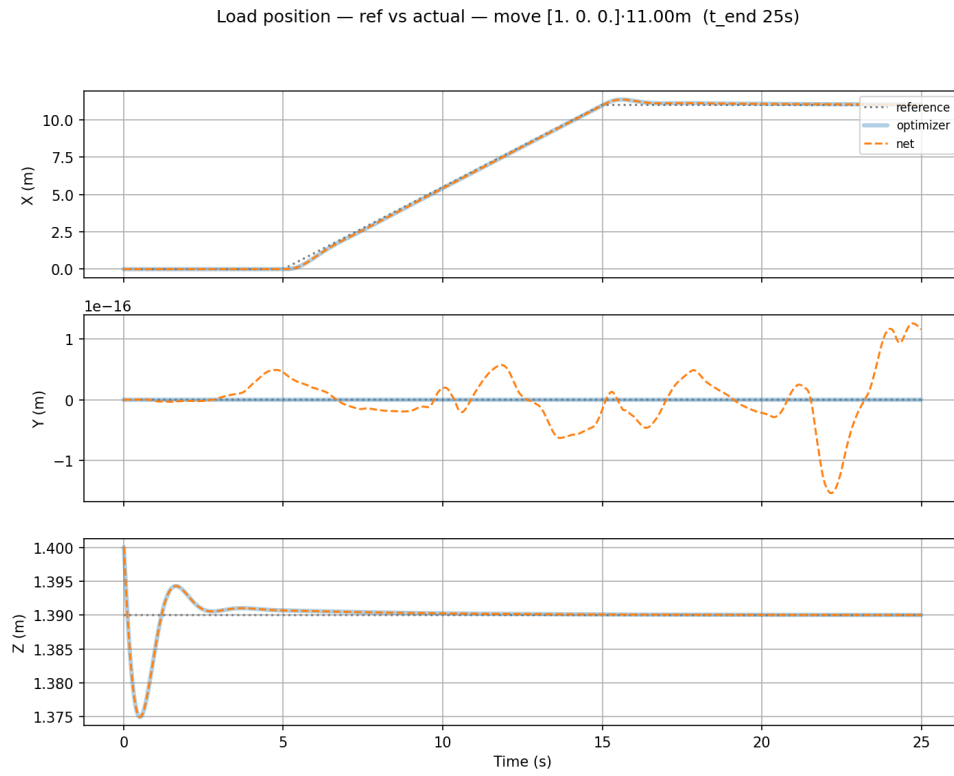


Figure 5.10: Load position tracking ($X/Y/Z$): reference vs network and optimizer.

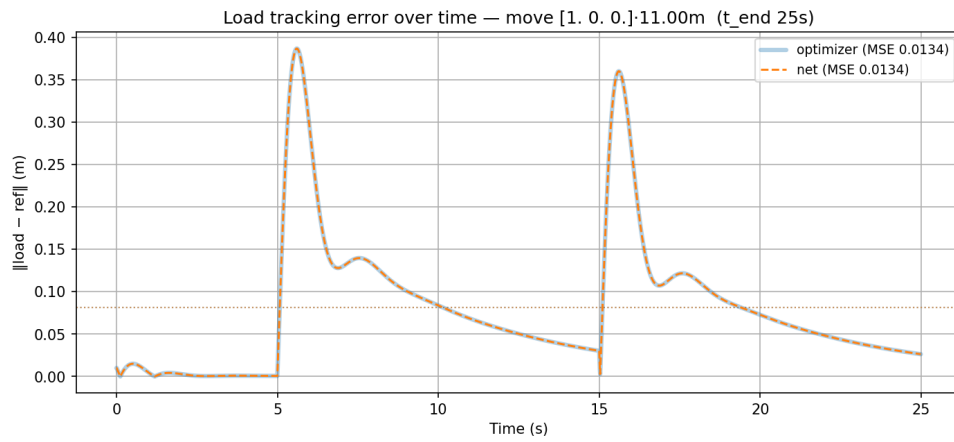


Figure 5.11: Load tracking error $\|load - ref\|$ over time: network vs optimizer.

The last two plots show that the load tracking is not changed by swapping the optimizer with the net (note that the scale on the Y-axis is 10^{-16}). This is expected as by design they do not affect the load tracking forces/wrench.

Top view (X-Y) — optimizer (solid) vs net (dashed) — move [1. 0. 0.]:11.00m (t_end 25s)

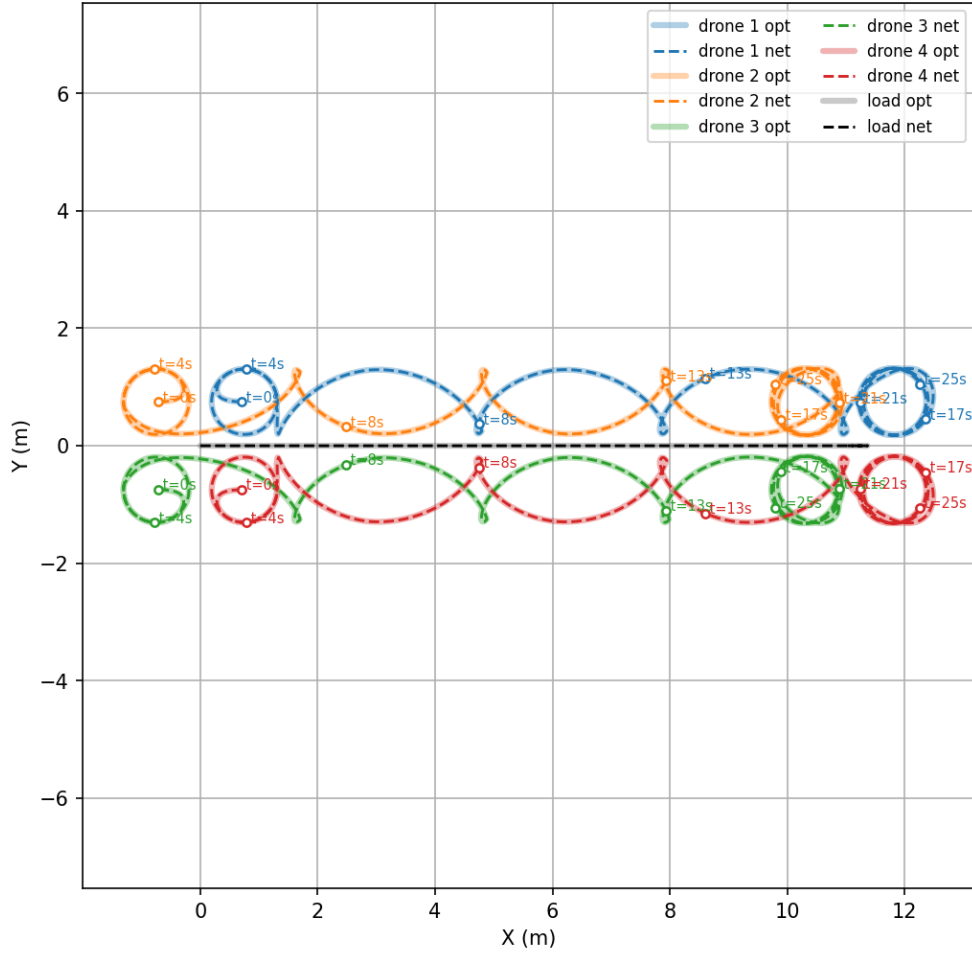


Figure 5.12: Carrier and load trajectories, top view (X-Y): network vs optimizer.

Drone XYZ trajectories — optimizer (solid) vs net (dashed) — move [1. 0. 0.]:11.00m (t_end 25s)

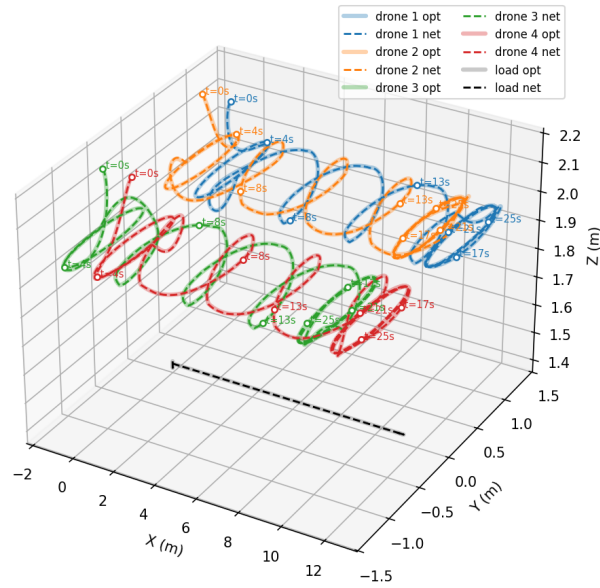


Figure 5.13: Carrier and load trajectories (X-Y-Z): network vs optimizer.

We can see that the carrier trajectories produced by the net overlap with the trajectories

produced by the CCS with the optimizer. This is expected since the λ 's are overlapping.

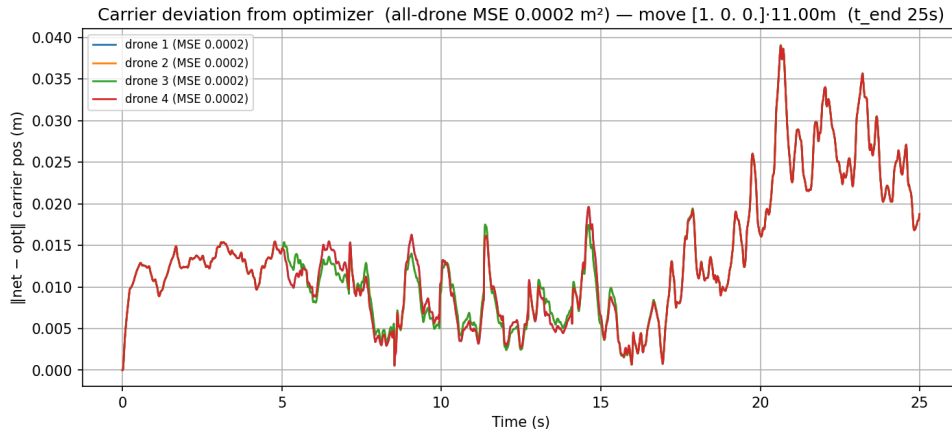


Figure 5.14: Per-carrier position deviation of the network from the optimizer over time.

We can see that the deviation in general is low and remains bounded below 0.04m (4cm) throughout the episode. While the error trends up from around $t = 15$ s to $t = 20$ s, it stabilizes and even trends down from $t = 20$ s to $t = 25$ s. We hypothesize that this behavior is due to the delayed effect of the historical λ 's in the network input. In other words, when A and ξ change mid-trajectory, it takes some time until the historical λ 's in the observation space are all produced by these new values. Therefore, when A and ξ change quickly, the deviation rises but then over time it stabilizes and goes back down after the net adjusts itself to the new values.

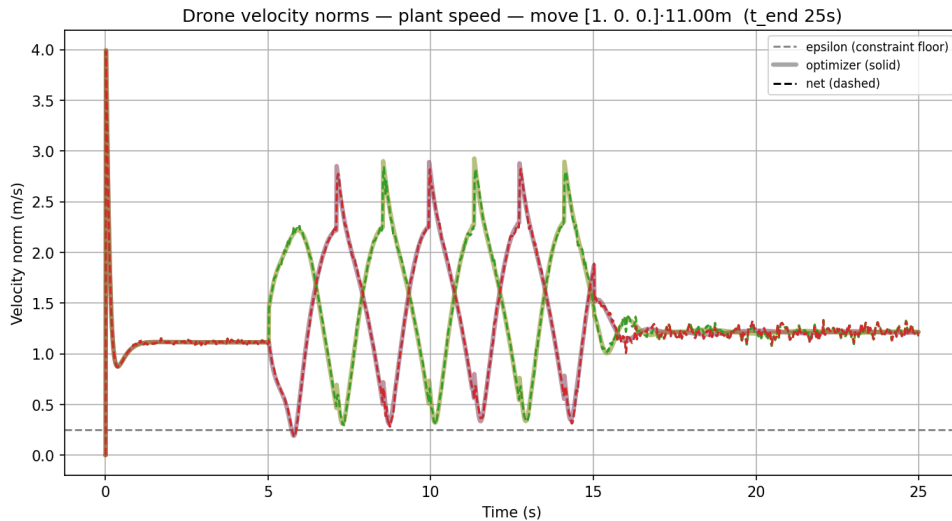


Figure 5.15: Carrier velocity norms: network vs optimizer, with the ϵ constraint floor.

We can see that the velocity norms overlap and that the net has learnt to respond to the minimum velocity violation in the exact same way as the optimizer. While it is a bit noiser especially towards the latter part of the episode, the noise is generally very small in magnitude.

Table 5.1: Performance of the DAgger-trained policy (network) on the single trajectory.

Metric	Network	Optimizer
Load tracking MSE (m^2)	0.01338	0.01338
Load tracking max error (m)	0.38677	0.38677
Carrier position deviation from optimizer, MSE (m^2)		
Drone 1	0.00023	--
Drone 2	0.00023	--
Drone 3	0.00023	--
Drone 4	0.00023	--
All drones	0.00023	--
λ deviation from optimizer, MSE		
Drone 1	0.00068	--
Drone 2	0.00076	--
Drone 3	0.00068	--
Drone 4	0.00076	--
All drones	0.00072	--

The load tracking is identical to the fifth decimal (and potentially beyond that but it serves no practical purpose). This is expected as explained earlier due to the structure and decomposition of the forces. The λ deviations mainly show up in the carrier positions' deviations but even then they remain minor.

We can see that we have succeeded in the task of overfitting one trajectory. Therefore it is reasonable to assume we have a potentially good state/input and network architecture. Thus, we can proceed with adding more trajectories and generalization.

5.1.2 Analysis of failure modes

Before we show generalization results, we show some of the failures encountered before obtaining these results. They are important to show because they guided the design choices that were discussed in the Methodology chapter.

We will focus on showing the λ 's since they are the direct output of the net and everything else is derived from it. In other words, if our policy is imitating the optimizer's λ 's correctly then we expect everything to be similar. Furthermore, if the λ 's are not identical, then the learning is not working and we need to analyze why at the λ 's level. Analyzing the other variables does not tell us what is the root cause of the problem.

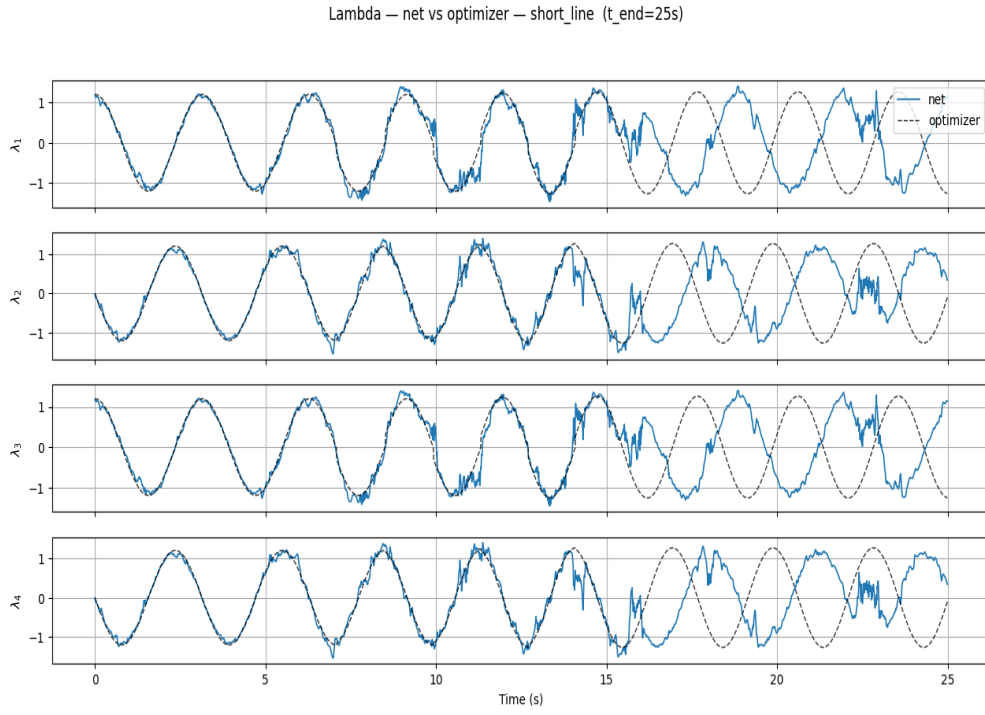


Figure 5.16: λ 's produced by the network versus the optimizer on a short-line trajectory: the network tracks the optimizer closely at first, then diverges as errors compound.

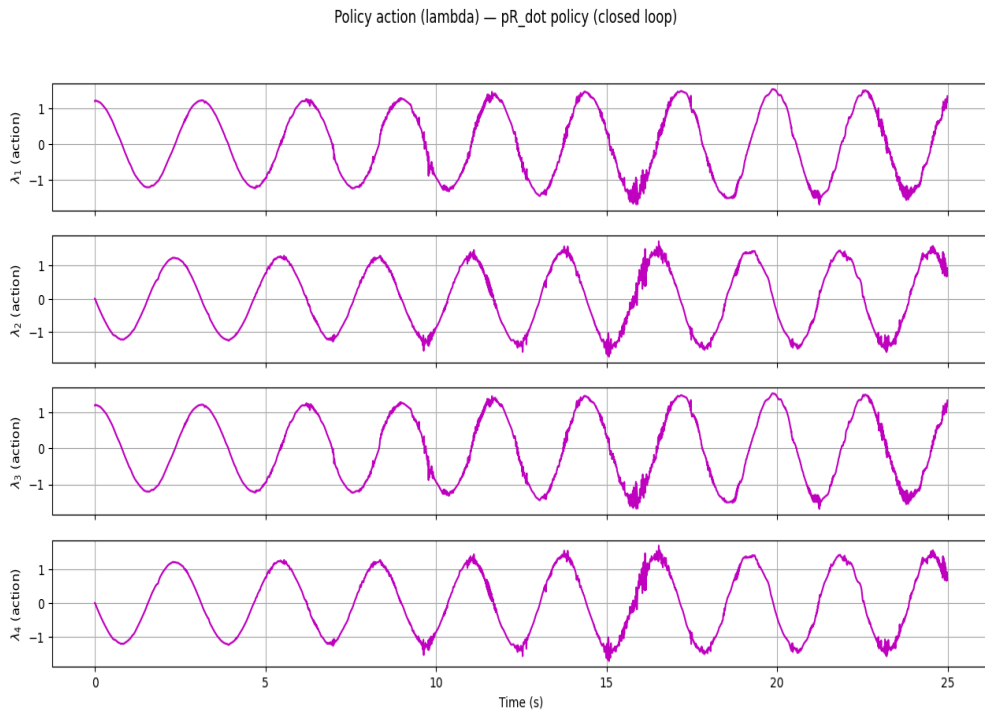


Figure 5.17: λ 's produced by a policy trained with $\dot{p}_R$ in the observation space, flown in closed loop: increasing high-frequency noise appears as the trajectory progresses.

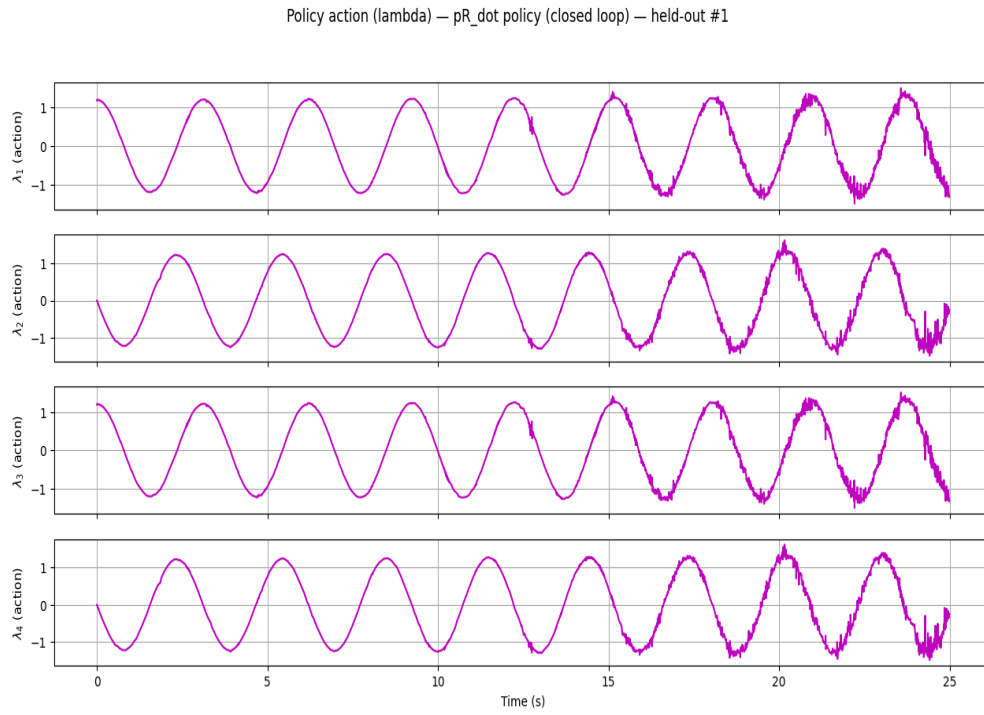


Figure 5.18: λ 's produced by the same $\dot{p}_R$ policy on a held-out trajectory: the same noise issue appears, showing it is not specific to a single trajectory.

5.1.3 Generalization

5.1.3.1 Generalization across time

Longer-time trajectory (move $[-0.06, -0.88, -0.47] \cdot 3 \text{ m}$, ramp, 50 s)

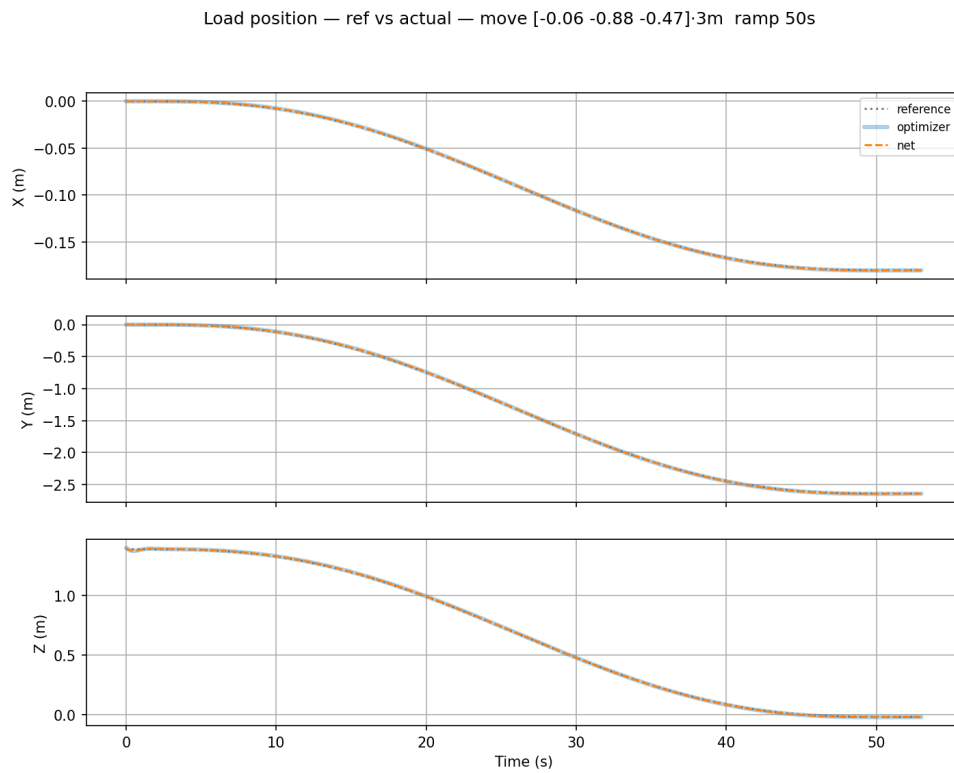


Figure 5.19: Load position: reference vs. actual.

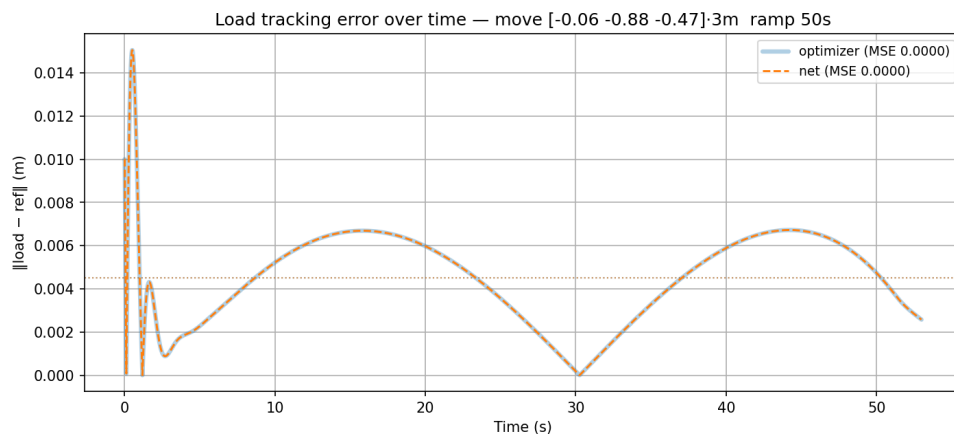


Figure 5.20: Load tracking error over time.

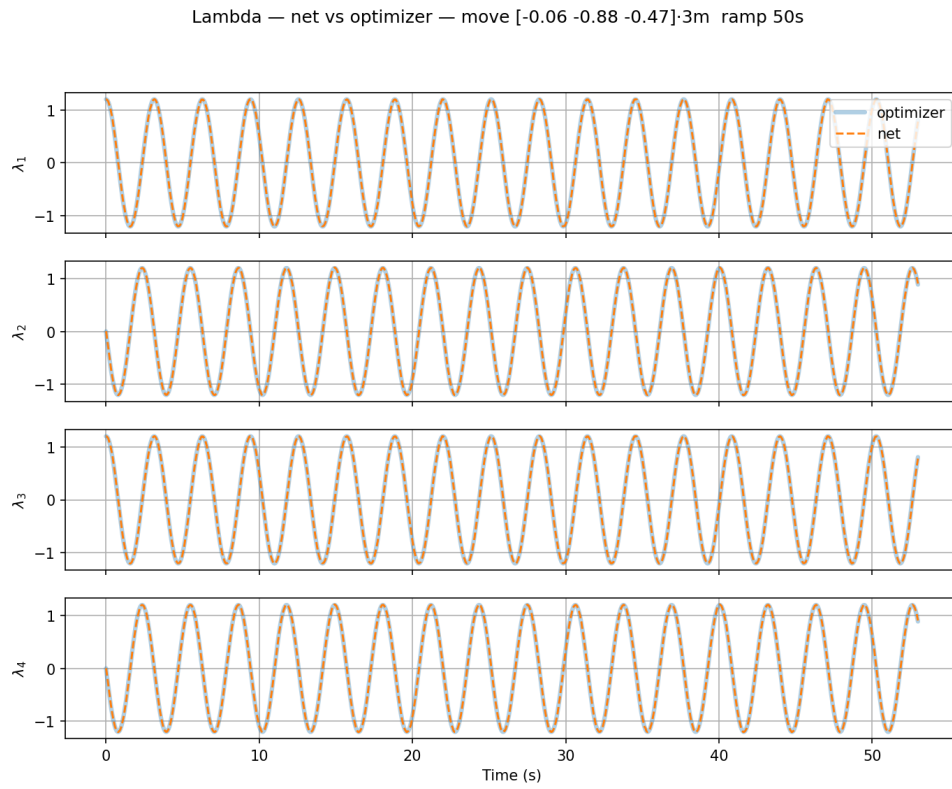


Figure 5.21: λ produced by the network vs. the optimizer.

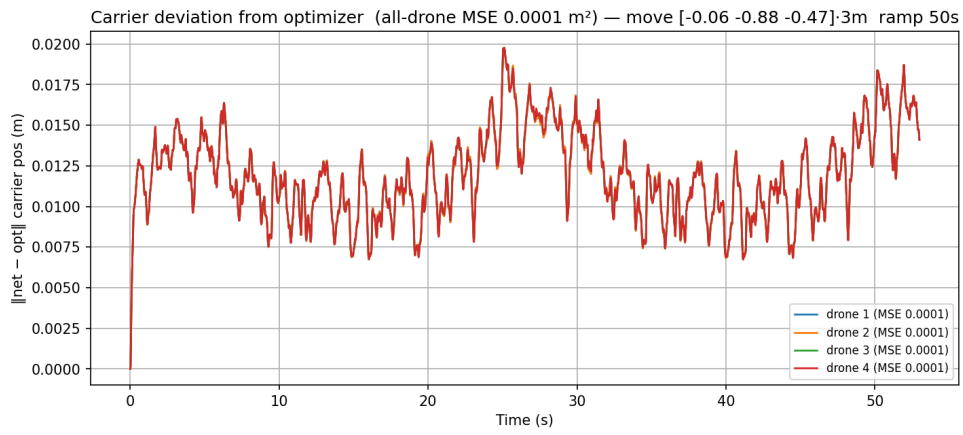


Figure 5.22: Carrier deviation from the optimizer.

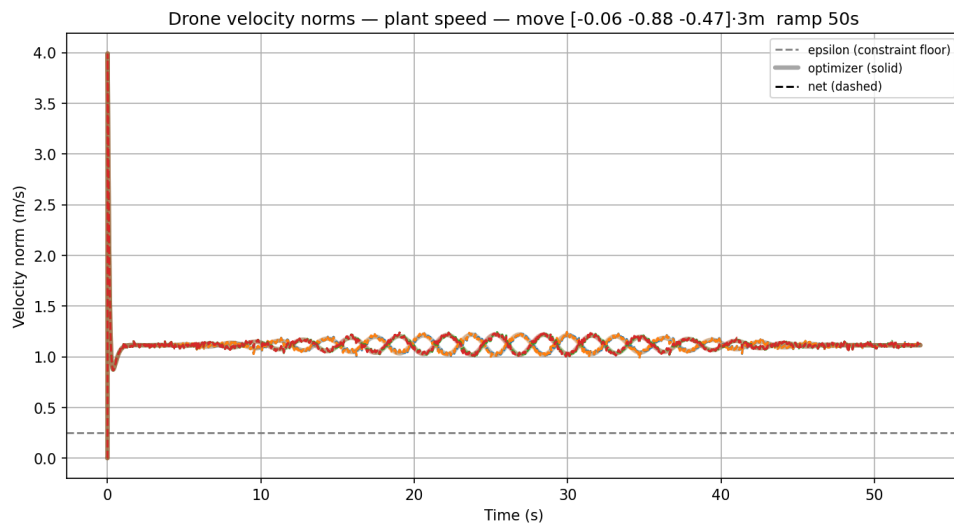


Figure 5.23: Drone velocity norms (plant speed).

Drone XYZ trajectories — optimizer (solid) vs net (dashed) — move [-0.06 -0.88 -0.47]·3m ramp 50s

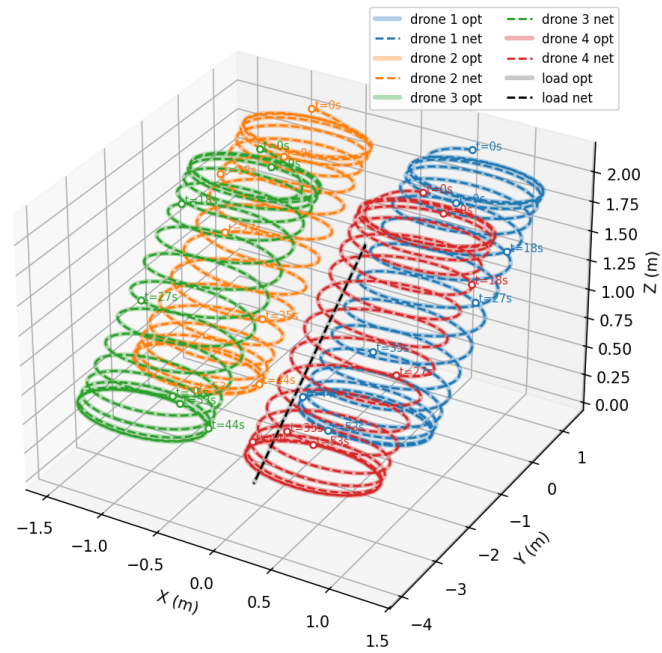


Figure 5.24: Drone XYZ trajectories: optimizer (solid) vs. net (dashed).

Table 5.2: Performance of the network on the longer-time generalization trajectory.

Metric	Network	Optimizer
Load tracking MSE (m^2)	0.00002	0.00002
Load tracking max error (m)	0.01505	0.01505
Carrier deviation from optimizer, MSE (m^2)		
Drone 1	0.00015	--
Drone 2	0.00015	--
Drone 3	0.00015	--
Drone 4	0.00015	--
All drones	0.00015	--
λ deviation from optimizer, MSE		
Drone 1	0.00035	--
Drone 2	0.00038	--
Drone 3	0.00035	--
Drone 4	0.00038	--
All drones	0.00037	--

Shorter-time trajectory (move $[-0.06, -0.88, -0.47] \cdot 3 \text{ m}$, ramp, 20 s)

Load position — ref vs actual — move $[-0.06 -0.88 -0.47] \cdot 3\text{m}$ ramp 20s

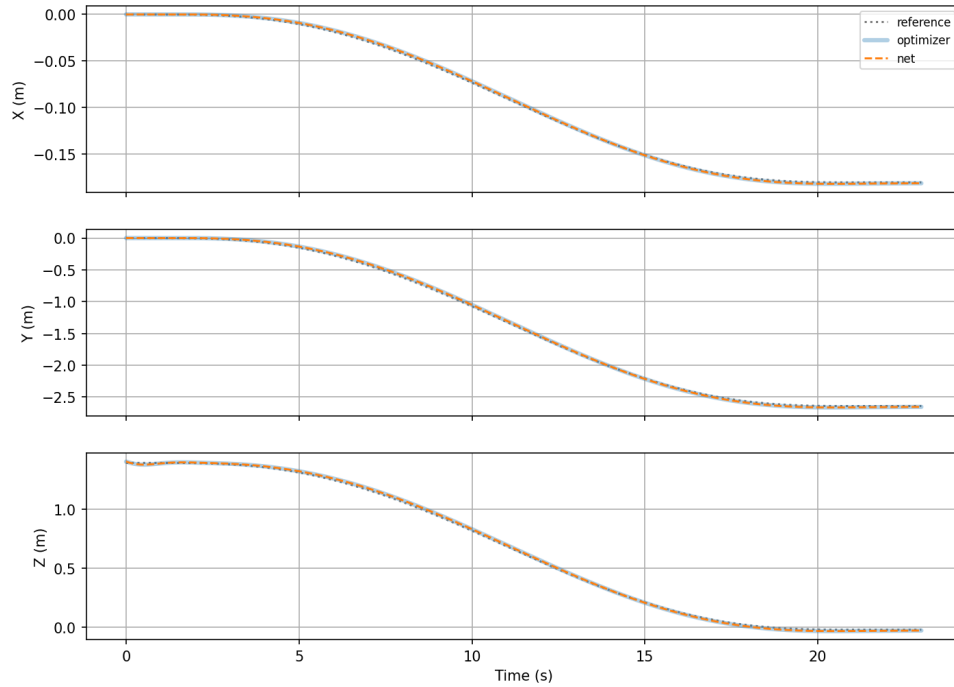


Figure 5.25: Load position: reference vs. actual.

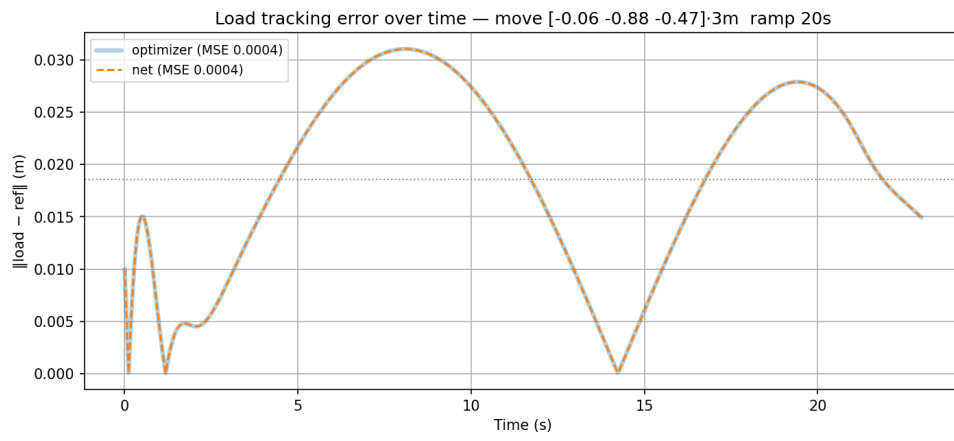


Figure 5.26: Load tracking error over time.

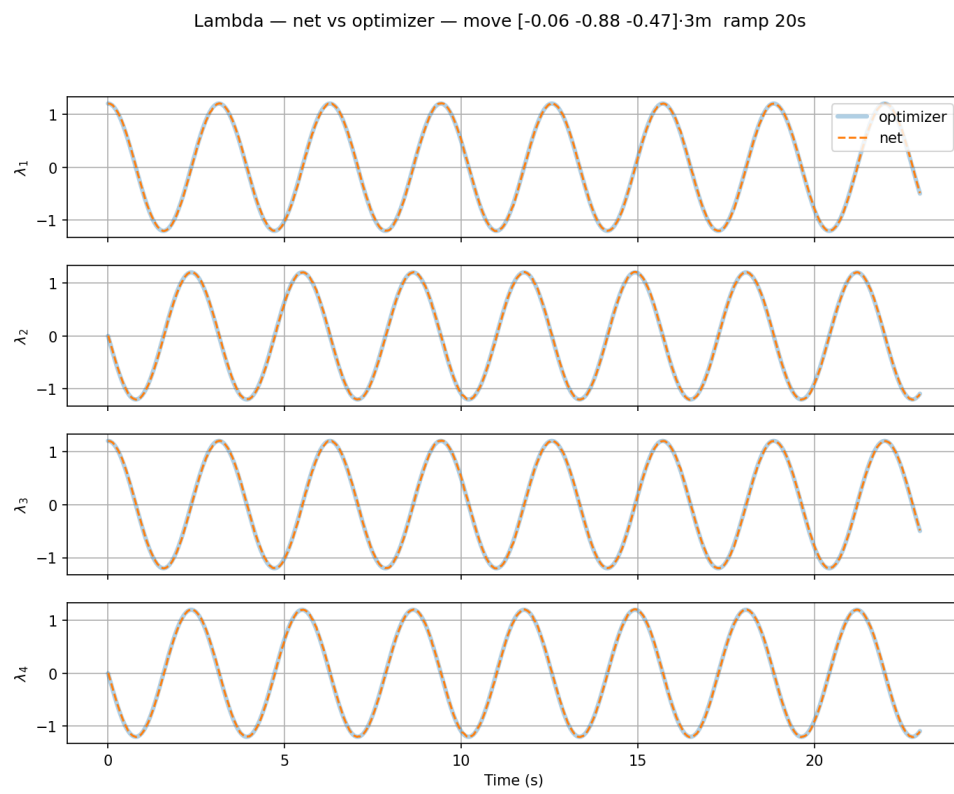


Figure 5.27: λ produced by the network vs. the optimizer.

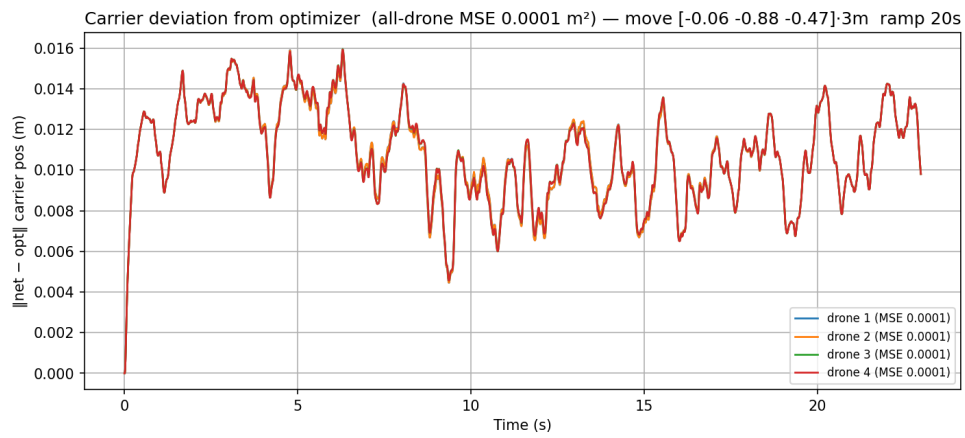


Figure 5.28: Carrier deviation from the optimizer.

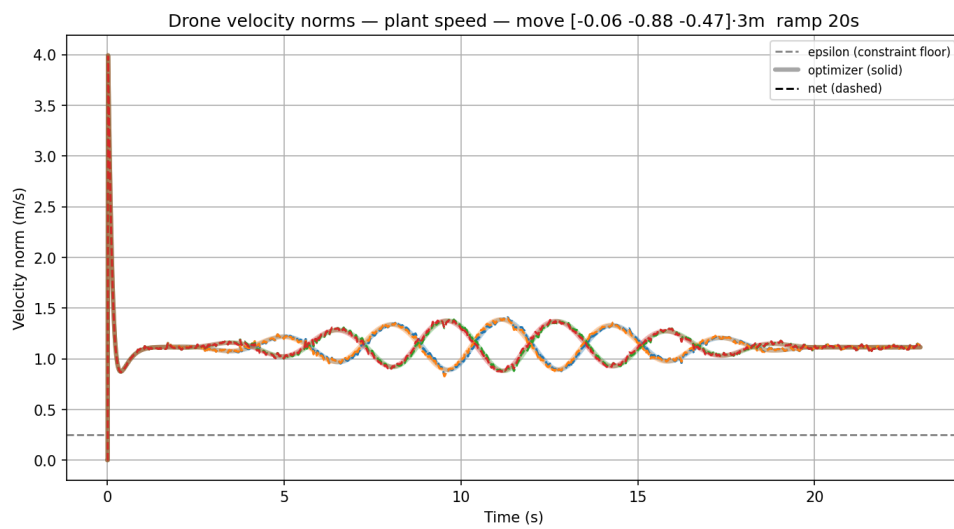


Figure 5.29: Drone velocity norms (plant speed).

Drone XYZ trajectories — optimizer (solid) vs net (dashed) — move $[-0.06 -0.88 -0.47] \cdot 3\text{m}$ ramp 20s

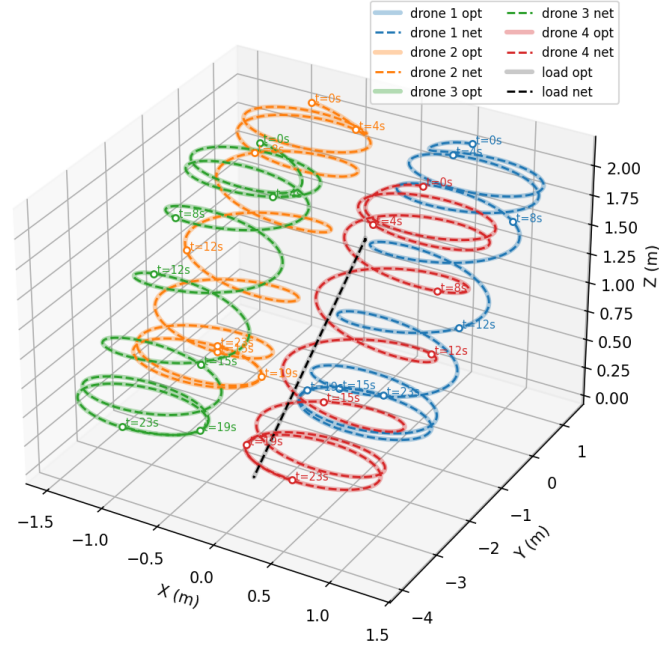


Figure 5.30: Drone XYZ trajectories: optimizer (solid) vs. net (dashed).

Table 5.3: Performance of the network on the shorter-time generalization trajectory.

Metric	Network	Optimizer
Load tracking MSE (m^2)	0.00043	0.00043
Load tracking max error (m)	0.03105	0.03105
Carrier deviation from optimizer, MSE (m^2)		
Drone 1	0.00012	--
Drone 2	0.00012	--
Drone 3	0.00012	--
Drone 4	0.00012	--
All drones	0.00012	--
λ deviation from optimizer, MSE		
Drone 1	0.00030	--
Drone 2	0.00032	--
Drone 3	0.00030	--
Drone 4	0.00032	--
All drones	0.00031	--

5.1.3.2 Generalization across distances

Larger-magnitude trajectory (move $[-0.06, -0.88, -0.47] \cdot 15\text{ m}$, ramp, 70 s)

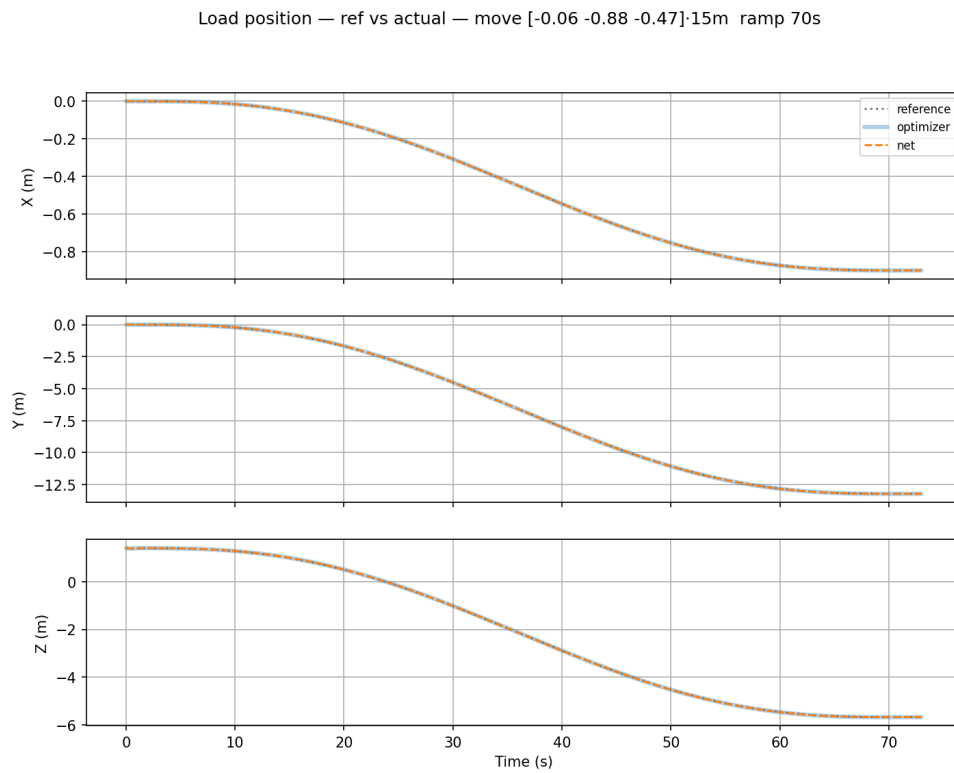


Figure 5.31: Load position: reference vs. actual.

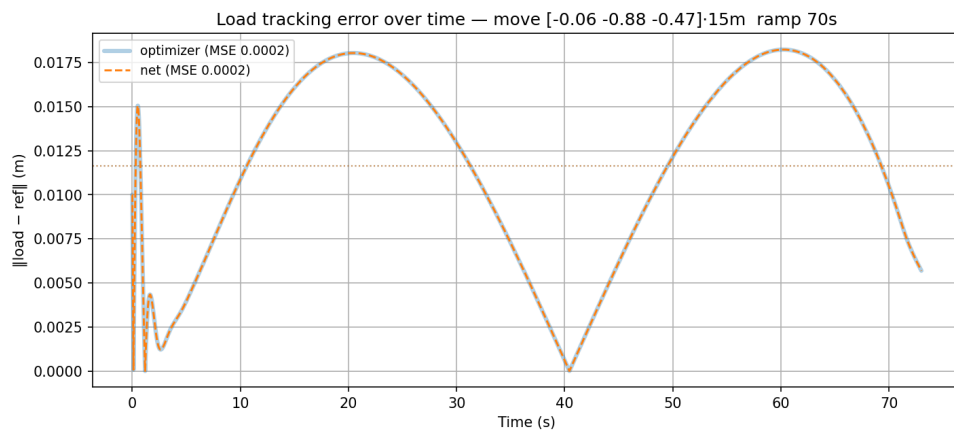


Figure 5.32: Load tracking error over time.

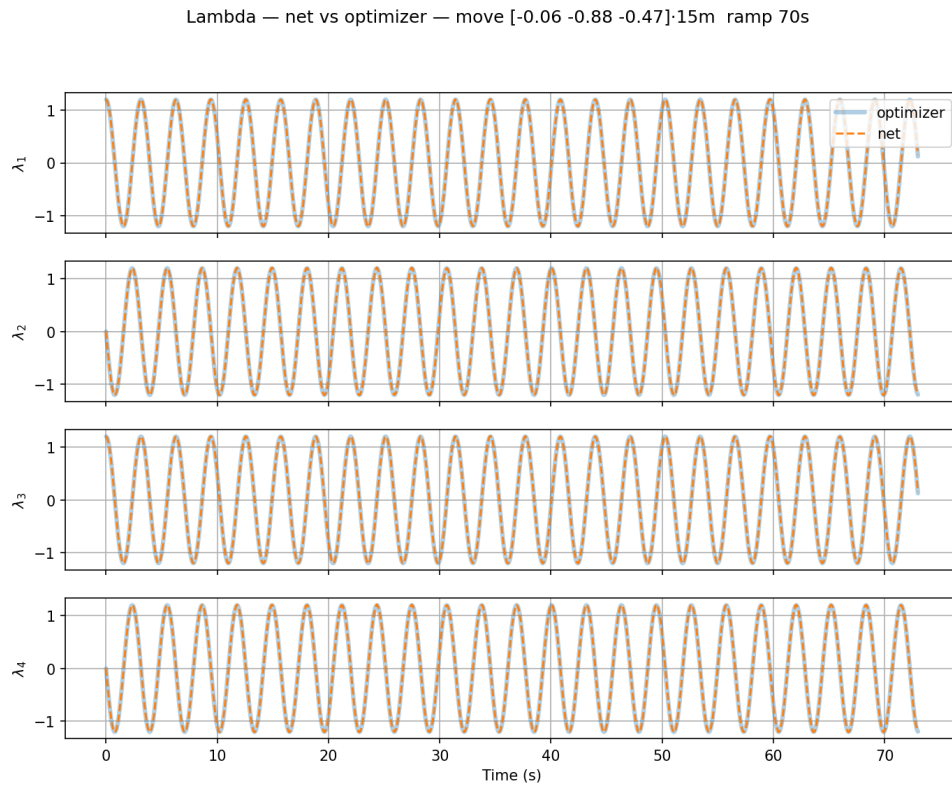


Figure 5.33: λ produced by the network vs. the optimizer.

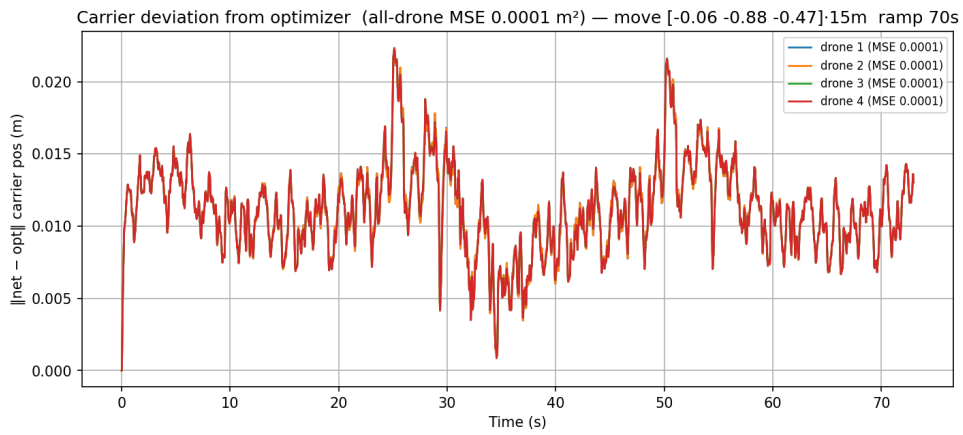


Figure 5.34: Carrier deviation from the optimizer.

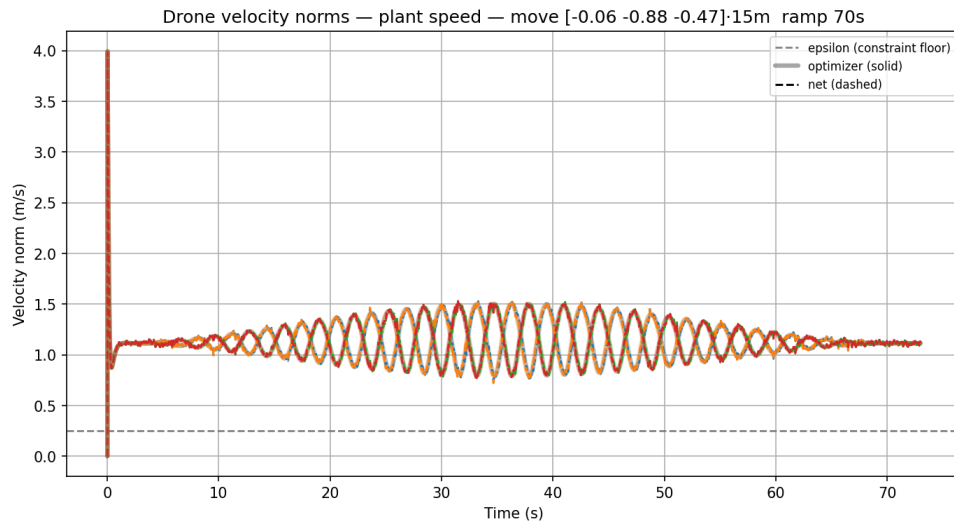


Figure 5.35: Drone velocity norms (plant speed).

Drone XYZ trajectories — optimizer (solid) vs net (dashed) — move [-0.06 -0.88 -0.47]·15m ramp 70s

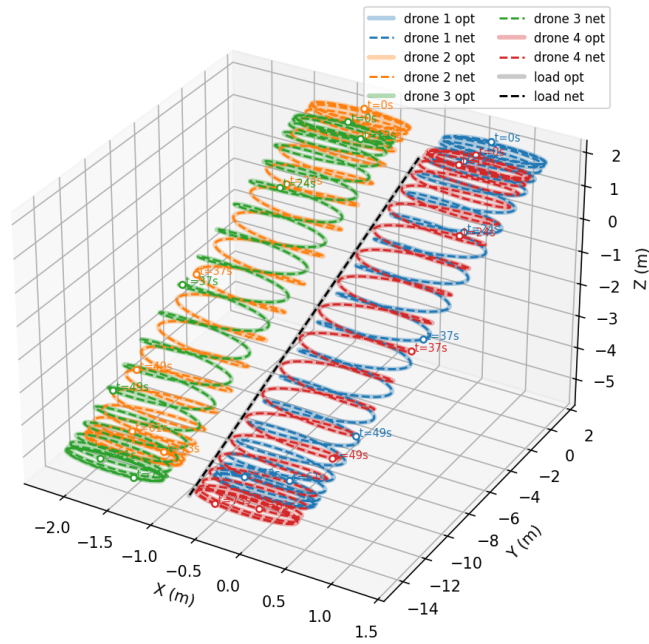


Figure 5.36: Drone XYZ trajectories: optimizer (solid) vs. net (dashed).

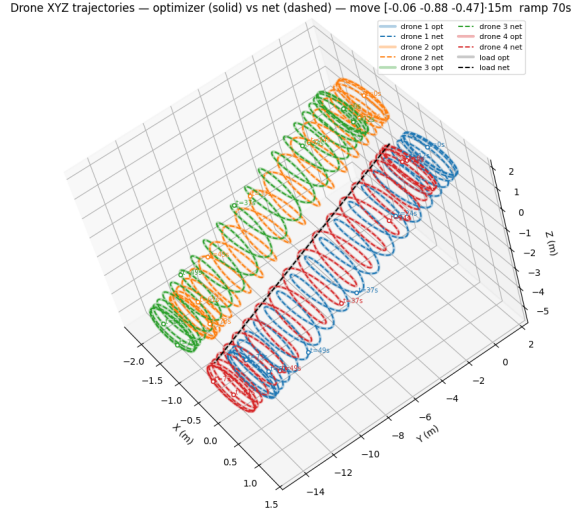


Figure 5.37: Drone XYZ trajectories, second point of view: optimizer (solid) vs. net (dashed).

Table 5.4: Performance of the network on the larger-magnitude generalization trajectory.

Metric	Network	Optimizer
Load tracking MSE (m ²)	0.00017	0.00017
Load tracking max error (m)	0.01824	0.01824
Carrier deviation from optimizer, MSE (m²)		
Drone 1	0.00013	--
Drone 2	0.00013	--
Drone 3	0.00013	--
Drone 4	0.00013	--
All drones	0.00013	--
λ deviation from optimizer, MSE		
Drone 1	0.00032	--
Drone 2	0.00036	--
Drone 3	0.00032	--
Drone 4	0.00036	--
All drones	0.00034	--

5.1.3.3 Generalization across directions

Direction 1 (move $[-0.5, -1, 0.5] \cdot 5$ m, ramp, 30 s)

Load position — ref vs actual — move [-0.5 -1. 0.5]-5m ramp 30s rpy r=10.1 p=-9.9 y=-0.9°

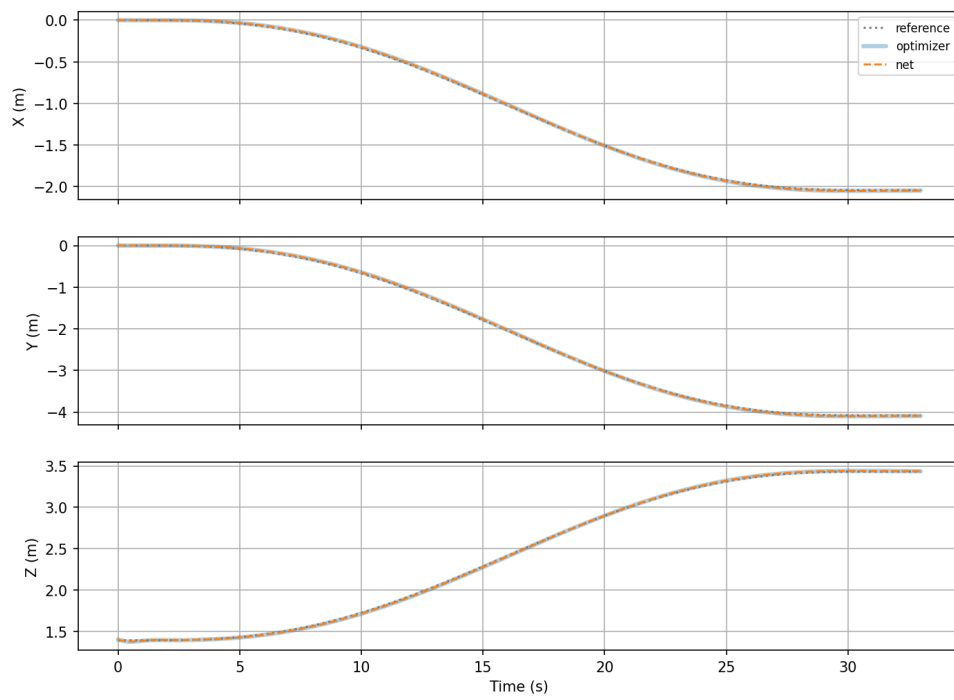


Figure 5.38: Load position: reference vs. actual.

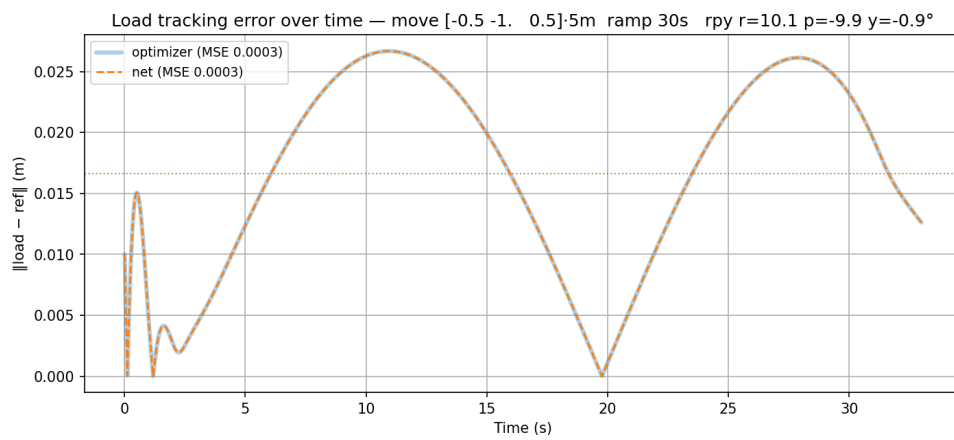


Figure 5.39: Load tracking error over time.

Load orientation — ref vs actual — move [-0.5 -1. 0.5]·5m ramp 30s rpy r=10.1 p=-9.9 y=-0.9°

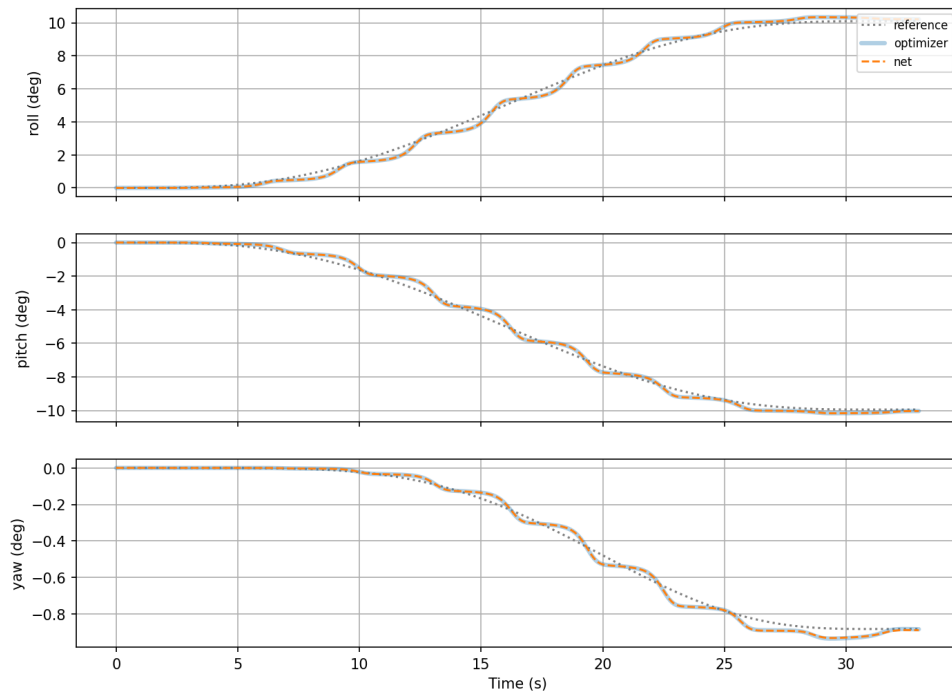


Figure 5.40: Load orientation: reference vs. actual.

Lambda — net vs optimizer — move [-0.5 -1. 0.5]·5m ramp 30s rpy r=10.1 p=-9.9 y=-0.9°

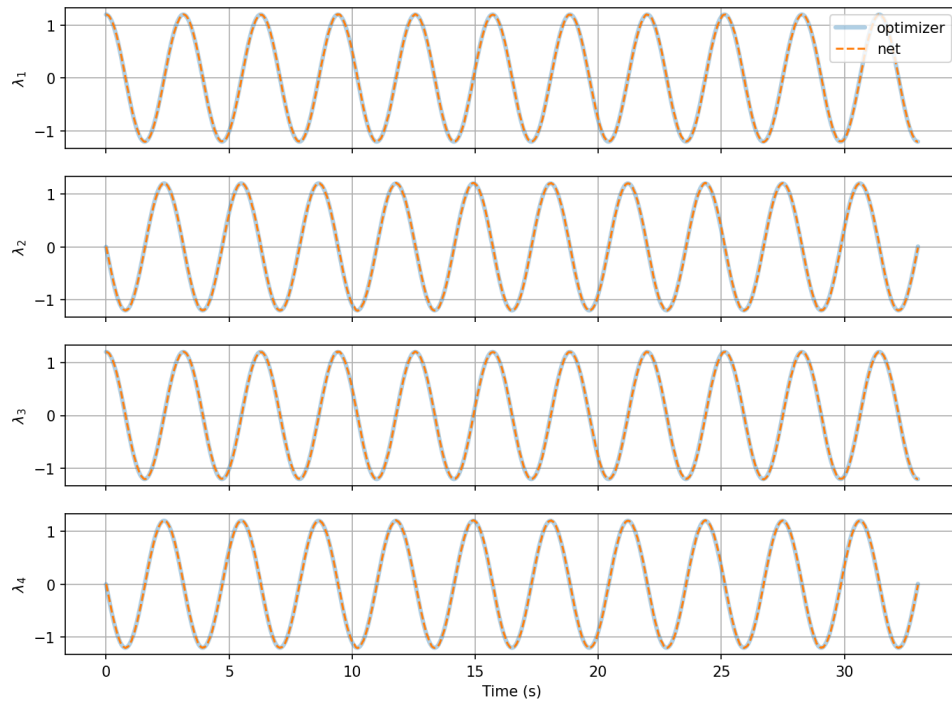


Figure 5.41: λ produced by the network vs. the optimizer.

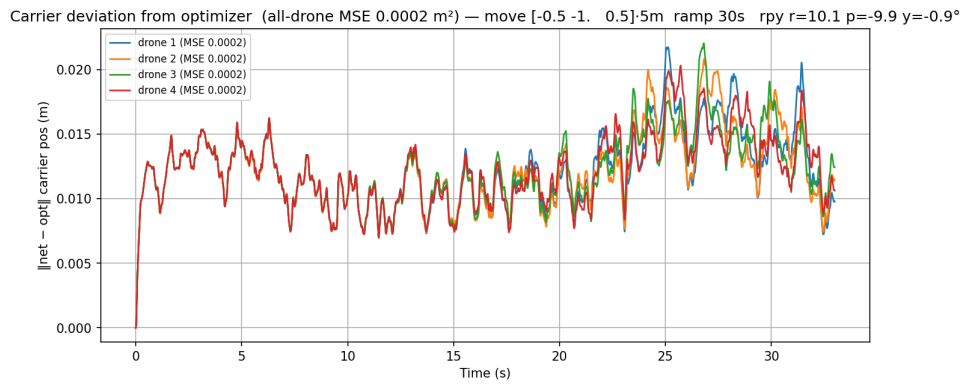


Figure 5.42: Carrier deviation from the optimizer.

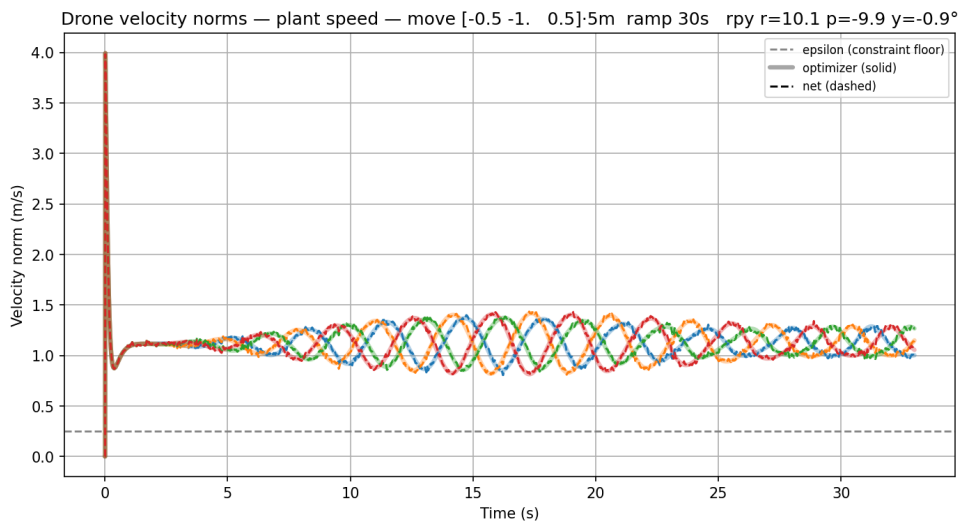


Figure 5.43: Drone velocity norms (plant speed).

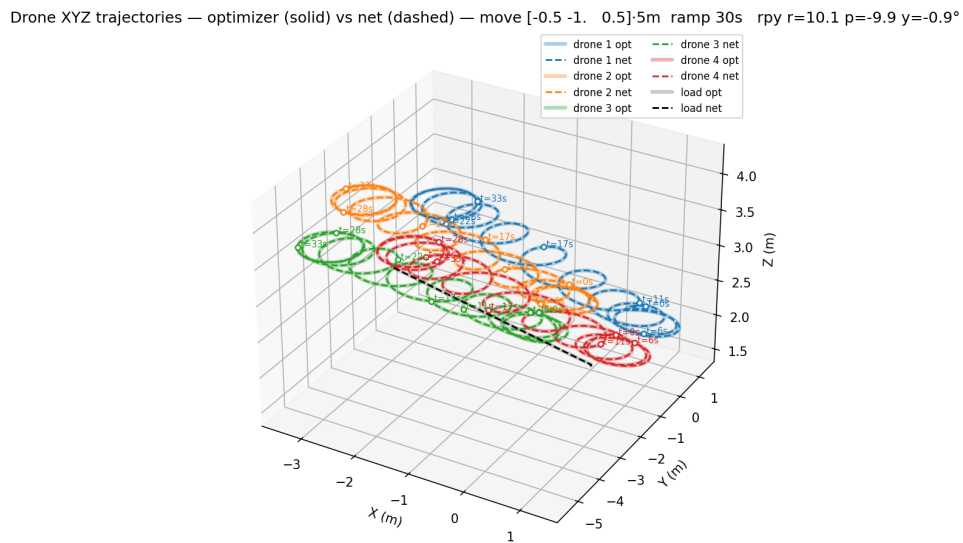


Figure 5.44: Drone XYZ trajectories: optimizer (solid) vs. net (dashed).

Table 5.5: Performance of the network on generalization direction 1.

Metric	Network	Optimizer
Load tracking MSE (m^2)	0.00034	0.00034
Load tracking max error (m)	0.02667	0.02667
Carrier deviation from optimizer, MSE (m^2)		
Drone 1	0.00016	--
Drone 2	0.00016	--
Drone 3	0.00016	--
Drone 4	0.00016	--
All drones	0.00016	--
λ deviation from optimizer, MSE		
Drone 1	0.00037	--
Drone 2	0.00042	--
Drone 3	0.00037	--
Drone 4	0.00042	--
All drones	0.00040	--

Direction 2 (move $[0.1, 0.1, -1] \cdot 5 \text{ m}$, ramp, 30 s)

Load position — ref vs actual — move [0.1 0.1 -1.]·5m ramp 30s rpy r=-5.0 p=5.0 y=-0.2°

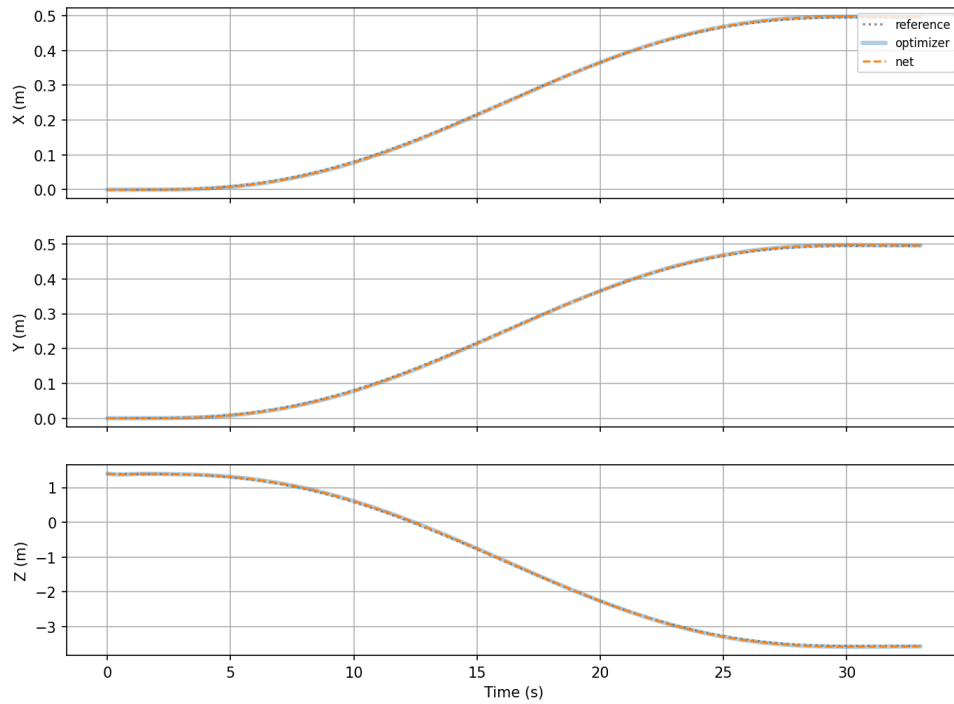


Figure 5.45: Load position: reference vs. actual.

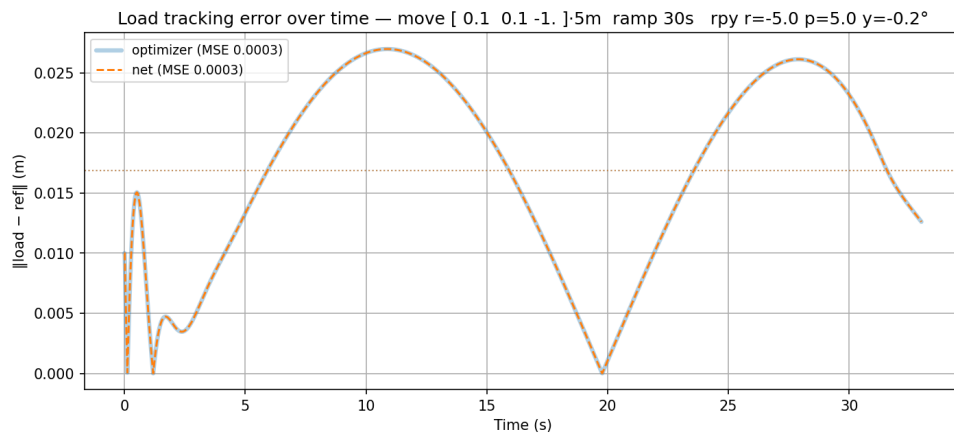


Figure 5.46: Load tracking error over time.

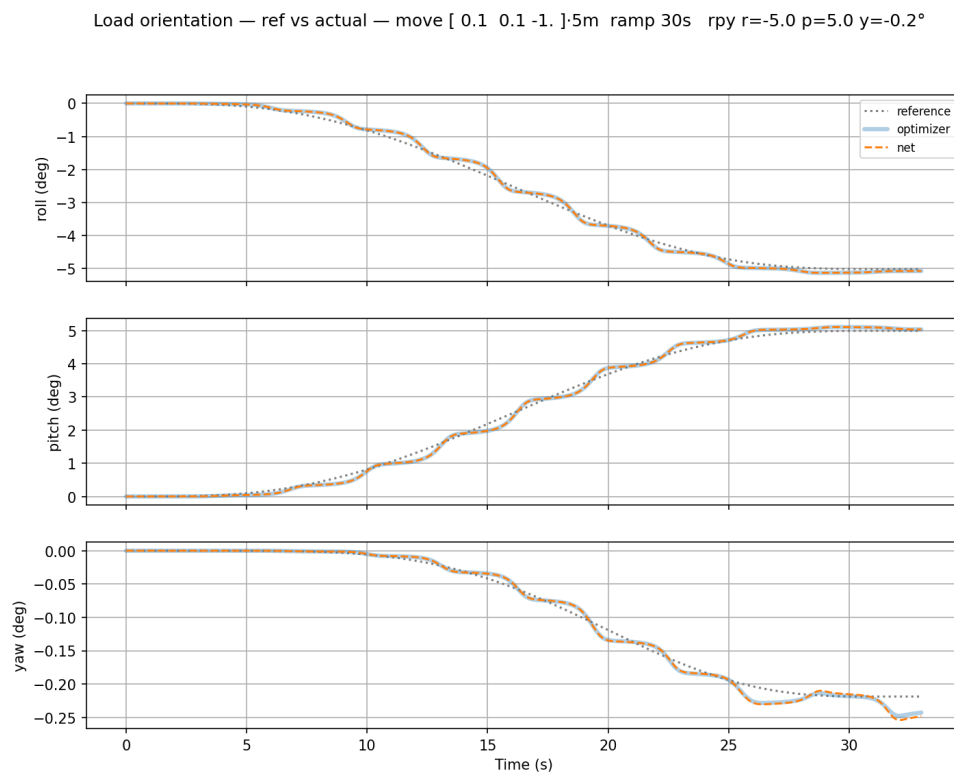


Figure 5.47: Load orientation: reference vs. actual.

Lambda — net vs optimizer — move [0.1 0.1 -1.]-5m ramp 30s rpy r=-5.0 p=5.0 y=-0.2°

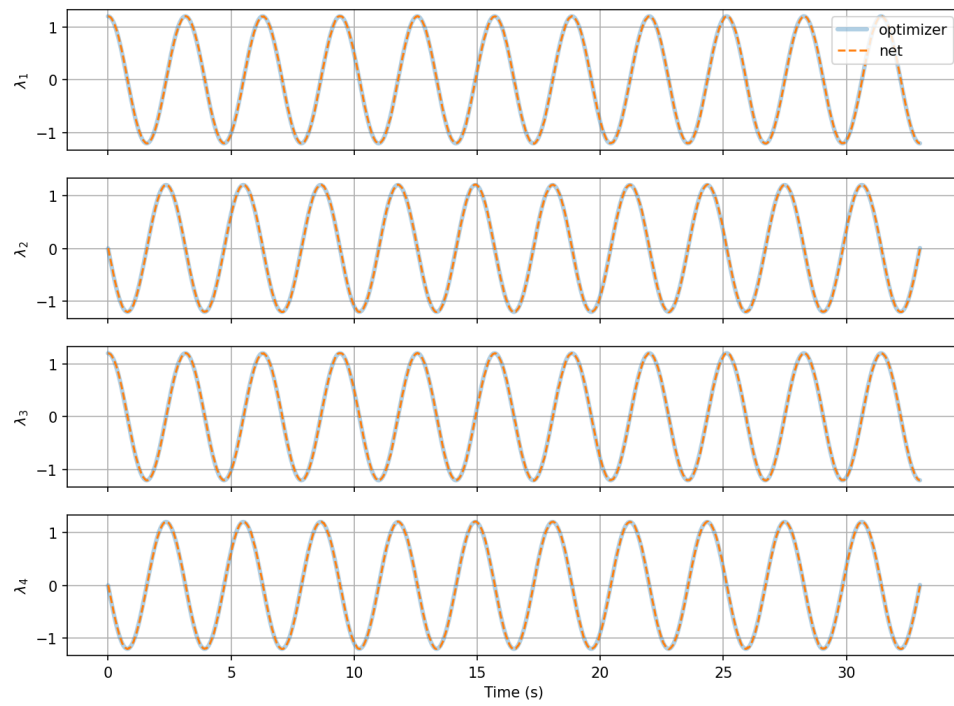


Figure 5.48: λ produced by the network vs. the optimizer.

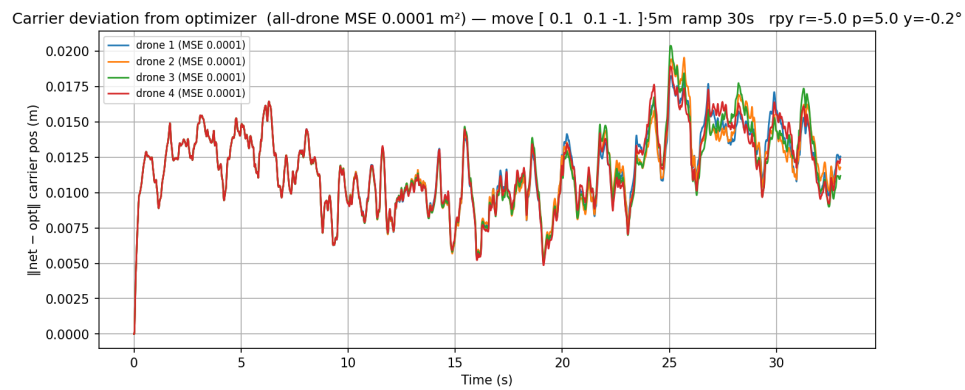


Figure 5.49: Carrier deviation from the optimizer.

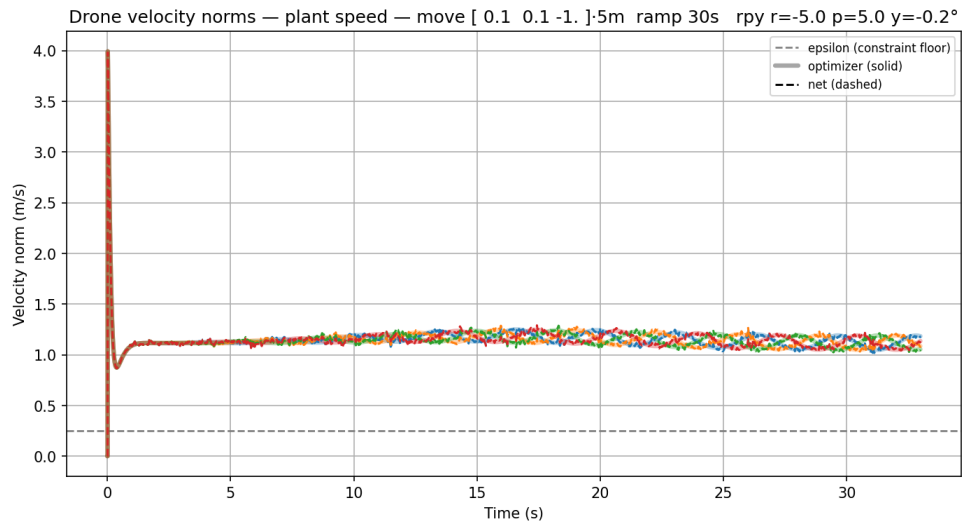


Figure 5.50: Drone velocity norms (plant speed).

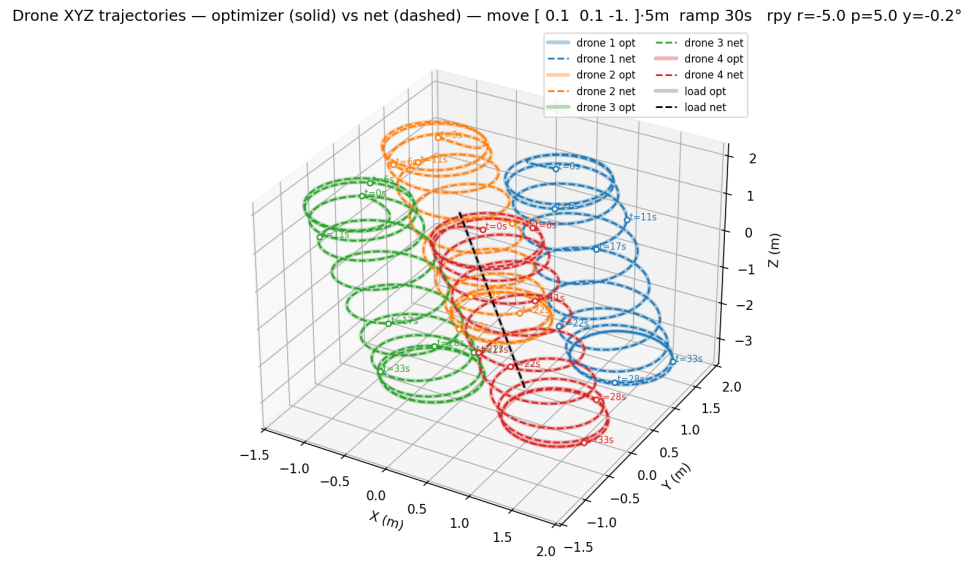


Figure 5.51: Drone XYZ trajectories: optimizer (solid) vs. net (dashed).

Table 5.6: Performance of the network on generalization direction 2.

Metric	Network	Optimizer
Load tracking MSE (m^2)	0.00035	0.00035
Load tracking max error (m)	0.02697	0.02697
Carrier deviation from optimizer, MSE (m^2)		
Drone 1	0.00014	--
Drone 2	0.00015	--
Drone 3	0.00015	--
Drone 4	0.00015	--
All drones	0.00015	--
λ deviation from optimizer, MSE		
Drone 1	0.00034	--
Drone 2	0.00038	--
Drone 3	0.00034	--
Drone 4	0.00038	--
All drones	0.00036	--

Summary across generalization trajectories

Table 5.7: Average network performance across all five generalization trajectories (longer-time, shorter-time, larger-magnitude, direction 1, direction 2).

Metric	Average (all drones)
Carrier deviation from optimizer, MSE (m^2)	0.00014
λ deviation from optimizer, MSE	0.00036

5.2 Phase Two : Reinforcement Learning

5.2.1 Single trajectory

5.2.2 Analysis of failure modes

5.2.3 Generalization

5.2.3.1 Generalization across time

5.2.3.2 Generalization across distances

5.2.3.3 Generalization across directions

5.2.3.4 History-Augmented Observation Space

5.2.3.5 Centralized Controller Policy Guidance experiment (using Supervised Learning)

5.2.3.6 Behavior Cloning (BC) of multiple policies

CHAPTER 6

Conclusion and Future Work

References

[1] E. N. Barmounakis, E. I. Vlahogianni, and J. C. Golias. Unmanned aerial aircraft systems for transportation engineering: Current practice and future challenges.

[2] citation needed

[3],[4] old papers and sofia's work

[5] Tancik et al. 2020, fourier time encoding

[6] [Ross et al., 2011]

[7] tu delft marl paper

[8] ar1 paper noise citation

[9] check out : Jerath, K et al (2018), "Bridging the gap between sensor noise modeling and sensor characterization," Or Besada-Portas et al. (2011) for delays

[10] An Empirical Model of Large-Batch Training

[11] Gradient Surgery for Multi-Task Learning" (PCGrad)